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Vibrationally adiabatic models for reactive tunneling

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The approximation of vibrational adiabaticity in curvilinear natural collision coordinates is investigated for tunneling in three-atom collinear reactions. A validity criterion is derived which limits the adiabatic approximation to systems with small reaction-path curvature. A general formalism is developed for systems which satisfy this criterion. A one-dimensional Schrödinger equation is proposed which is sufficiently flexible so as to be adaptable to many different models of tunneling. We present three new methods for including reaction-path curvature effects on multidimensional tunneling in reactive systems: a method based on a quantum mechanical vibrational average (VA) over degrees of freedom transverse to the minimum-energy path; a method (called DA for dynamical-path vibrational-average) that includes internal centrifugal effects in the description of the transverse vibrational motion (in this method the vibrational average is approximated as a quantal vibrational average about the dynamical path along which the Born–Oppenheimer force cancels the internal centrifugal force); and a semiclassical optical potential (SOP) method based on the Feshbach formalism translated into an adiabatic representation with reaction-path curvature providing the coupling mechanism between the explicit and implicit spaces. These models are compared, both formally and numerically, to each other and to four other methods that have been proposed previously, including the small-curvature (SC) approximation that we have proposed in a recent communication. The VA and SOP methods are shown to provide generalizations of phase average (PA) and second-order (SO) methods proposed earlier by Miller and co-workers. The difference is that vibrations are treated quantum mechanically in the VA and SOP methods but classically and harmonically in the PA and SO methods; the quantum mechanical methods have the advantage that anharmonicity can be included more straightforwardly. The DA, SO, and SOP methods, although they include internal centrifugal effects more fully than the VA and PA methods, do not offer significant improvement in accuracy. The numerical results clearly support the physical interpretation of the collapse of the vibrational wave function about a least-action path. The most successful methods are the Marcus–Coltrin path (MCP) and SC approximations. These methods, especially the SC approximation because it is more general, are recommended for future applications.

I. INTRODUCTION

One-dimensional potential-barrier models are very useful for the treatments of tunneling in chemical reactions. Following the early work of Wigner¹ and Eckart,² many such models have been proposed. Recently, sophisticated multichannel quantal calculations have become feasible for sufficiently simple systems.³ Nevertheless, interest in one-dimensional models has not waned. For example, recent advances in transition state theory and generalized transition state theory have shown that thermally averaged quantal transmission coefficients calculated from a one-dimensional model can be used to calculate accurate thermal rate constants.^{4,5} In such applications, the one-dimensional model allows a tremendous reduction in computation time over a complete quantal calculation when the latter is possible. One-dimensional transmission coefficients can also be used to incorporate quantal effects into trajectory calculations.⁵ Furthermore, for most three-atom systems and all systems with four or more atoms multichannel calculations are not presently feasible, while a one-dimensional treatment remains tractable.⁶ Another reason for the continuing interest in simplified one-dimensional models is their intuitive value. Indeed, a physical interpretation of the reactive process is usually built into the model from the beginning, and numerical comparisons to accurate quantal results reflect on the validity of that interpretation.

The most successful one-dimensional models proposed so far have been based in one way or another on the assumption of adiabaticity in natural collision coordinates.^{7–14} In these methods, the rotational and vibrational coordinates are removed from explicit consideration, leaving an effective Schrödinger equation with a potential that is a function of a single reaction coordinate. The technical difficulty that distinguishes reactive adiabaticity from the Born–Oppenheimer variety is the need to handle the curvilinear nature of the reaction coordinate. There are two reasons why reaction-path curvature affects the reaction process. First, because of curvature, the reaction-path length is dependent on the internal vibrational phase. This effect can be treated either by the use of a coordinate-dependent effective mass or equivalently by a redefinition of the reaction-coordinate metric. A second result of reaction-path curvature is centrifugal distortion of the internal states. In this paper, we are primarily interested in the behavior of the reactive systems in the classically forbidden tunneling region. There are, broadly speaking, two rival ways to extend the vibrational adiabaticity concept to tunneling regions. The first asserts that the internal vibrational-rotational wave function retains its general structure as it moves through the classically forbidden region. According to this view, an effective one-dimensional Schrödinger equation is obtained by averaging the total Hamiltonian

over the adiabatic internal state. A second interpretation holds that the internal wave function effectively collapses about a minimum-imaginary-action path. In this case, the transmission probability is obtained by calculating the action integral along this dominant path. The purpose of this paper is threefold. First, we provide a unified formalism which can encompass both of these interpretations and their plethora of individual expressions for small- and medium-curvature systems. Second, in terms of this formalism, we develop several original models and compare their dominant physical characteristics to each other and to many theories already presented in the literature. Third, we provide numerical comparisons to exact two-mathematical-dimensional results to make a practical conclusion concerning the best physical extension of vibrational adiabaticity to tunneling regions. For simplicity, we shall restrict the explicit development to collinear three-atom reactions. With one exception, the DA method (see below), all the models considered do have straightforward higher-dimensional extensions.

We conclude the introduction with a brief outline of the format we shall employ. In Sec. II, the general features of the Schrödinger equation in curvilinear natural collision coordinates are discussed, and a validity criterion for subsequent development is presented. In Sec. III, a model adiabatic Hamiltonian is developed, and we present a generic equation for motion along the reaction coordinate. This defining equation [Eqs. (35)–(39)] is used to provide a unified presentation of the different adiabatic models in Sec. V. Section IV describes the semiclassical methods used for obtaining the transmission probability using the one-dimensional model adiabatic Hamiltonian. Section V is a presentation of seven different one-dimensional models. For convenience we shall employ a consistent set of abbreviations for these methods, namely, PA (phase average), VA (vibrational average), DA (dynamical-path vibrational-average), MCP (Marcus-Coltrin path), SC (small curvature), SO (second order), and SOP (semiclassical optical potential). The SC method has been described briefly in a preliminary communication,¹⁴ and the VA, SOP, and parts of the DA model are original to this paper; the remaining theories already exist in the literature. The harmonic versions of the models are presented in Secs. VA to VG. In Sec. VH, the anharmonic generalizations are presented. Section VI contains a comparison of the models on formal grounds. Sections VII and VIII present numerical results for collinear reactions using the seven models and WKB theory, and Sec. IX gives a few results for three-dimensional reactions. Section X is the conclusion.

II. COORDINATES

Throughout this article we assume the Born–Oppenheimer separation of electronic and nuclear motions and, except in Sec. IX, collinear A + BC collisions. Thus, after separating out the center of mass motion, the wave function depends on two coordinates. The basis for obtaining a one-dimensional description of the reaction process is the assumption of vibrational adiabaticity in curvilinear natural collision coordinates. We define a generic set

of collision coordinates in which t is the reaction coordinate (specified below), and v is the transverse vibrational coordinate. Vibrational adiabaticity means that the vibrational quantum number n for the v mode is conserved. In actuality, it is only in the asymptotic regions $t \rightarrow \pm\infty$ that adiabaticity becomes exact. One attempts to choose a coordinate system which, as much as possible, preserves adiabaticity in the interaction region while properly representing free motion in the asymptotic regions. Two coordinate systems are considered here. The first is based on the minimum-energy path (MEP) reaction coordinate, which is defined as the path of steepest descent on the Born–Oppenheimer potential surface starting from the saddle point. In particular, t measures arc length along the minimum-energy path through a mass-scaled coordinate system in which the reduced mass for each direction has the same value μ . [Such a coordinate system will be called a scaled diagonalizing (SD) system.] In the second coordinate system, the reaction coordinate is chosen to be the dynamical path (DP) along which the centripetal force is balanced by the restoring force of the potential within the framework of a particular adiabatic separation of the Hamiltonian. The DP reaction coordinate is located on the convex side of the MEP when the system is in a classically allowed region (i.e., when the energy is greater than the adiabatic potential). For classically forbidden regions the centrifugal force is negative, and the DP is on the concave side of the MEP. For both the MEP and DP coordinates, v is linear in the SD coordinates mentioned above, and in these coordinates it is normal to t at $v=0$; v is defined here as positive on the large-internuclear-distance side of the reaction coordinate. Both the MEP and DP coordinate systems are special cases of natural collision coordinates, which have been discussed extensively elsewhere.^{11,15–18}

Natural collision coordinates exhibit multivaluedness over an unbounded region of space. A formal consequence of this condition is the necessity to impose a set of constraints on the total wave function. For example, if (t, v) and (t', v') correspond to the same point in space then

$$\psi(t, v) = \psi(t', v'). \quad (1)$$

If these constraints are not imposed in solving the Schrödinger equation, there is no guarantee that the solution will resemble the physical solution. Multivaluedness generally presents no problems in the asymptotic region since the curvature of the reaction path becomes zero and the reactant and product valleys become well separated. Consequently, matrix elements of the interaction Hamiltonian between reasonable adiabatic basis functions will not be affected by the multivaluedness. For the multivalued region which lies on the concave side of the reaction coordinate in the interaction region, the situation is more serious. There one finds several sets of ordered pairs with comparable values of v , and one can expect some sort of overlap between vibrations taking place at different positions on the reaction coordinate. To obtain reliable results without the imposition of constraints, the multivalued region should be located sufficiently far from the re-

action coordinate so that the wave function there is forced by the potential to be small. If this is the case, then even though the wave function is incorrect in the multivalued region, the probability density there is so small that physical observables can be accurately computed from the unconstrained Schrödinger equation. By making several simplifying assumptions, we can determine a general criterion for the validity of the unconstrained Schrödinger equation. Define the curvature of the reaction coordinate in an SD coordinate system as $\kappa(t)$. Then the multivalued region includes at least the region beyond the locus of points $\{t, 1/\kappa(t)\}$. A requirement for the unconstrained Schrödinger equation to be physically realistic is that the probability density at this boundary is small compared to that on the reaction coordinate, i.e.,

$$\left| \frac{\psi(t, v = \frac{1}{\kappa(t)})}{\max \psi(t, v)} \right|^2 \ll 1, \quad (2)$$

where the maximum in the denominator is taken over all v at fixed t . We estimate this ratio as follows. We assume a vibrationally adiabatic wave function of the form

$$\psi(t, v) = f_n(t) \phi_n(v; t), \quad (3)$$

where $\phi_n(v; t)$ is the bound vibrational wave function which solves

$$\left[-\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial v^2} + V_2(v; t) - \epsilon_n(t) \right] \phi_n(v; t) = 0, \quad (4)$$

with

$$V_2(v; t) = V(t, v) - V(t, v=0), \quad (5)$$

where $V(v, t)$ is the Born-Oppenheimer potential and $f_n(t)$ is an unbound reaction-coordinate wave function. Then the criterion becomes

$$\left| \frac{\phi_n(v = \frac{1}{\kappa(t)}; t)}{\max \phi_n(v; t)} \right|^2 \ll 1, \quad (6)$$

where the maximum in the denominator is taken over all v at fixed n and t . For $n=0$ and a harmonic vibrational potential this becomes

$$\exp[-\mu \omega(t)/\hbar \kappa^2(t)] \ll 1, \quad (7)$$

with $\omega(t)$ being the harmonic frequency. The validity of the unconstrained Schrödinger equation in reaction coordinates thus depends on the smallness of the parameter

$$\frac{\hbar \kappa^2(t)}{\mu \omega(t)} \equiv a^2(t). \quad (8a)$$

We shall find that the parameter $a(t)$ is a central quantity in our subsequent developments. In particular, we note that $a^2(t)$ can also be written as

$$a^2(t) = \kappa^2(t) v_s^2(t), \quad (8b)$$

where $v_s(t)$ is the outer classical turning point of the ground state. The parameter $a(t)$ [or $a(s)$ in the notation of the next paragraph] will occur in the form (8a) or (8b) for harmonic potentials throughout Secs. V A to V G or in the form (8b) in Sec. V H.

One more item of notation: (t, v) refer to the generic coordinates, which may be either the MEP or the DP set. When we need to distinguish between these choices we use (s, u) for the MEP set and (ρ, w) for the DP set.

III. DEVELOPMENT OF A MODEL ADIABATIC HAMILTONIAN

In this section, a generic model equation for motion along the reaction coordinate is developed. We begin with the full Schrödinger equation for the collinear A + BC reaction, make a vibrationally adiabatic approximation, and eliminate terms which are either unphysical or computationally insignificant.

With the restriction of the last section, i.e., small $a^2(t)$, we can write an unconstrained total system wave function as

$$\psi(t, v) = \sum_n f_n(t) \phi_n(v; t). \quad (9)$$

Here $f_n(t)$ should satisfy the correct scattering boundary conditions. To specify the vibrational states and determine the defining differential equations for $f_n(t)$, it is desirable to perform an adiabatic separation of the Hamiltonian into bound and unbound components. Expressed in the generic collision coordinates, the total Schrödinger equation becomes¹⁵

$$(H - E) \psi(t, v) = 0 \quad (10a)$$

or

$$\left[-\frac{\hbar^2}{2\mu} \frac{1}{\eta} \frac{\partial}{\partial t} \frac{1}{\eta} \frac{\partial}{\partial t} - \frac{\hbar^2}{2\mu} \frac{1}{\eta} \frac{\partial}{\partial v} \eta \frac{\partial}{\partial v} + V(t, v) - E \right] \psi(t, v) = 0, \quad (10b)$$

where

$$\eta = 1 - \kappa(t)v \quad (11)$$

and $V(t, v)$ is the Born-Oppenheimer potential. We can make a general adiabatic separation by requiring the vibrational states to satisfy

$$[H_{vib} - \epsilon_n(t)] \phi_n(v; t) = 0, \quad (12a)$$

where

$$H_{vib} = -\frac{\hbar^2}{2\mu} \frac{1}{\eta} \frac{\partial}{\partial v} \eta \frac{\partial}{\partial v} + V_2(v; t) \quad (12b)$$

and $V_2(v; t)$ is a yet unspecified potential which may be energy dependent. The kinetic energy operator in Eq. (12) can be simplified by the replacement¹⁵

$$\phi_n(v; t) = \frac{1}{\eta^{1/2}} \chi_n(v; t). \quad (13)$$

Then Eq. (12) becomes

$$\left[-\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial v^2} + V_2(v; t) - \frac{\hbar^2 \kappa^2(t)}{8\mu \eta^2} - \epsilon_n(t) \right] \chi_n(v; t) = 0. \quad (14)$$

A set of close coupling equations can be obtained for the $f_n(t)$ by projecting out the v motion:

$$\langle \phi_j | H - E | \psi \rangle = 0, \quad (15a)$$

i.e.,

$$\sum_n \left[\int_{-\infty}^{1/\kappa(t)} J(t, v) \left\{ \frac{1}{\eta^{1/2}} \chi_n^*(v; t) \left[-\frac{\hbar^2}{2\mu} \frac{1}{\eta} \frac{\partial}{\partial t} \frac{1}{\eta} \frac{\partial}{\partial t} + V_1(t, v) \right] \right\} \right. \\ \left. \times \frac{1}{\eta^{1/2}} \chi_n(v; t) dv + \delta_{jn} [\epsilon_j(t) - E] \right] f_n(t) = 0, \\ j = 1, 2, \dots, N, \quad (15b)$$

where

$$V_1(t, v) = V(t, v) - V_2(v; t) \quad (16)$$

and the Jacobian is

$$J(t, v) = \eta. \quad (17)$$

In the adiabatic approximation, we neglect the nonadiabatic coupling of $f_n(t)$ to other basis elements. The defining equation is then

$$\left[\int_{-\infty}^{1/\kappa(t)} \eta^{1/2} \chi_n^*(v; t) \left[-\frac{\hbar^2}{2\mu} \frac{1}{\eta} \frac{\partial}{\partial t} \frac{1}{\eta} \frac{\partial}{\partial t} + V_1(t, v) \right] \right. \\ \left. \times \frac{1}{\eta^{1/2}} \chi_n(v; t) dv + \epsilon_n(t) - E \right] f_n(t) = 0. \quad (18)$$

The vibrationally averaged kinetic energy operator of Eq. (18) is quite complicated due to the action of $(\partial/\partial t)$ on η and $\chi_n(v, t)$. A popular approximation is to ignore terms which arise in this way. To investigate the significance of these terms, henceforth referred to as nonadiabatic derivative terms, we shall first calculate their effect explicitly for two test cases using particular choices for the coordinate system and for $V_2(v; t)$. We use the MEP coordinate system and define $V_2(u; s)$ and $\chi_n(u; s)$ by the following equations:

$$V_2(u; s) = V(s, u) - V(s, u = 0), \quad (19a)$$

$$\left[-\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial u^2} + V_2(u; s) - \epsilon_n(s) \right] \chi_n(u; s) = 0. \quad (19b)$$

We consider the ground vibrational state for this calculation. Equation (18) becomes

$$\left[-\frac{\hbar^2}{2\mu_{\text{eff}}(s, 0)} \frac{\partial^2}{\partial s^2} + \hbar B_0(s) \frac{\partial}{\partial s} + C_0(s) - E \right] f_0(s) = 0, \quad (20)$$

where

$$\frac{1}{\mu_{\text{eff}}(s, 0)} = \frac{A_0(s)}{\mu}, \quad (21a)$$

$$A_0(s) = \int_{-\infty}^{1/\kappa(s)} \frac{|\chi_0(u; s)|^2}{\eta^2} du, \quad (21b)$$

$$B_0(s) = \left(-\frac{\hbar}{2\mu} \right) \left[2\kappa' \int_{-\infty}^{1/\kappa(s)} \frac{u}{\eta^3} |\chi_0(u; s)|^2 du \right. \\ \left. + 2 \int_{-\infty}^{1/\kappa(s)} \frac{\chi_0^*(u; s) \chi_0'(u; s)}{\eta^2} du \right], \quad (22)$$

$$C_0(s) = \left(-\frac{\hbar^2}{2\mu} \right) \left[\int_{-\infty}^{1/\kappa(s)} \left(\frac{5}{4} \frac{(\kappa' u)^2}{\eta^4} + \frac{\kappa'' u}{2\eta^3} \right) |\chi_0(u; s)|^2 du \right. \\ \left. + \int_{-\infty}^{1/\kappa(s)} \frac{\chi_0^*(u; s) \chi_0''(u; s)}{\eta^2} du + 2 \int_{-\infty}^{1/\kappa(s)} \frac{\kappa' u}{\eta^3} \chi_0^*(u; s) \chi_0'(u; s) du \right] \\ + V^{\text{MEP}}(s) - \frac{\kappa^2(s) \hbar^2}{8\mu} A_0(s) + \epsilon_0(s), \quad (23)$$

where

$$V^{\text{MEP}}(s) = V(s, u = 0), \quad (24)$$

and primes denote derivatives with respect to s . For

convenience the arguments (s) or $(u; s)$ of the derivative quantities are not shown explicitly. The arguments (s, u) of η are also implicit. Following Bowman,¹⁹ the upper limit for the integration in Eq. (21) is set equal to $1/\kappa(s)$. To calculate $A_0(s)$ we use a method that will be described in detail later, the harmonic PA method. The harmonic PA method treats the expectation values as classical phase averages with a harmonic approximation to the vibrational potential. The extension of this method to calculate $B_0(s)$ and $C_0(s)$ is described in an appendix. The local vibrational zero-point energy $\epsilon_0(s)$ in $C_0(s)$ is, however, found by a more accurate Morse approximation (Morse approximation I as described elsewhere⁹). Later in this paper we consider several other methods for performing the expectation values in Eqs. (21)–(23); these yield somewhat different numerical results, but the main conclusions of this section are not sensitive to these differences since only qualitative deductions are made. With the harmonic PA method we have

$$\frac{1}{\mu_{\text{eff}}^{\text{PA}}(s, 0)} = \frac{1}{\mu} A_0^{\text{PA}}(s), \quad (25a)$$

$$A_0^{\text{PA}}(s) = \frac{1}{[1 - \alpha^2(s)]^{3/2}}, \quad (25b)$$

$$B_0(s) = -\frac{\hbar}{2\mu} \left\{ \frac{3\kappa'\kappa(s) A_0^{5/3}(s)}{\alpha^2(s)} \right. \\ \left. + 3 \frac{\alpha'}{\alpha(s)} [A_0(s) - A_0^{5/3}(s)] \right\}, \quad (26)$$

$$C_0(s) = V_a^G(s) - \frac{\kappa^2(s) \hbar^2}{8\mu_{\text{eff}}^{\text{PA}}(s, 0)} + \left(-\frac{\hbar^2}{2\mu} \right) \left\{ \frac{5}{8} \frac{(\kappa')^2 A_0^{5/3}(s)}{\alpha^2(s)} \right. \\ \times \left[1 + \frac{5\kappa^2(s)}{\alpha^2(s)} A_0^{2/3}(s) \right] + \frac{3}{4} \frac{\kappa''\kappa(s)}{\alpha^2(s)} A_0^{5/3}(s) - \frac{\kappa(s)\kappa'\alpha'}{\alpha^3(s)} \\ \times \left[\frac{15}{2} A_0^{7/3}(s) - \frac{9}{2} A_0^{5/3}(s) \right] + \frac{3}{2} \frac{\alpha''}{\alpha(s)} [A_0(s) - A_0^{5/3}(s)] \\ \left. + \left[\frac{\alpha'}{\alpha(s)} \right]^2 \left[\frac{7}{4} A_0(s) - 6A_0^{5/3}(s) + \frac{15}{4} A_0^{7/3}(s) \right] \right\}, \quad (27)$$

where $A_0(s)$ in Eqs. (26) and (27) refers to $A_0^{\text{PA}}(s)$ in Eq. (25b),

$$V_a^G(s) = V^{\text{MEP}}(s) + \epsilon_0(s) \quad (28a)$$

and

$$\alpha(s) = \left[\frac{\mu \omega(s)}{\hbar} \right]^{1/2}, \quad (28b)$$

with $\omega(s)$ being the harmonic frequency.

The quantities appearing in Eqs. (25)–(28) were calculated for two test cases $\text{D} + \text{D}_2$ and $\text{D} + \text{H}_2$, using a realistic potential energy surface presented elsewhere.²⁰ The minimum-energy path for $\text{D} + \text{H}_2$ is shown in Fig. 1, and various aspects of the results for both isotopic cases are shown in Figs. 2 and 3. Parts b of these figures are comparisons showing the size of various nonadiabatic derivative terms in $C_0(s)$. We see that, because of the nonadiabatic derivative terms, $C_0(s)$ often differs from the zero-curvature adiabatic ground-state potential $V_a^G(s)$ by 1 kcal/mol or more. The chief source of this deviation is seen to be the κ' and κ'' terms. The α' and α'' terms have significantly

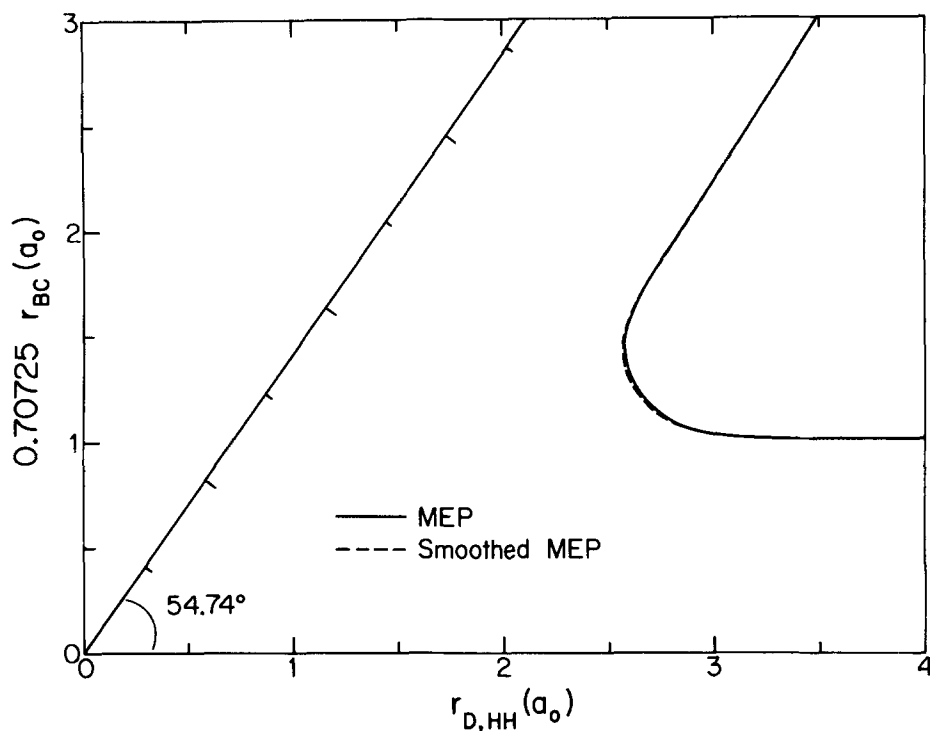


FIG. 1. Natural collision coordinates for the collinear $D+H_2$ reaction. The solid curve is the minimum energy path and the dashed curve is a smooth reaction path whose curvature is consistent with Eq. (32).

less effect. To show the influence of the nonadiabatic derivative terms of $B_0(s)$, we have made semiclassical calculations of the transmission probabilities. Using parabolic uniformization,²¹ we approximated the semiclassical reaction probability at energy E for the one-

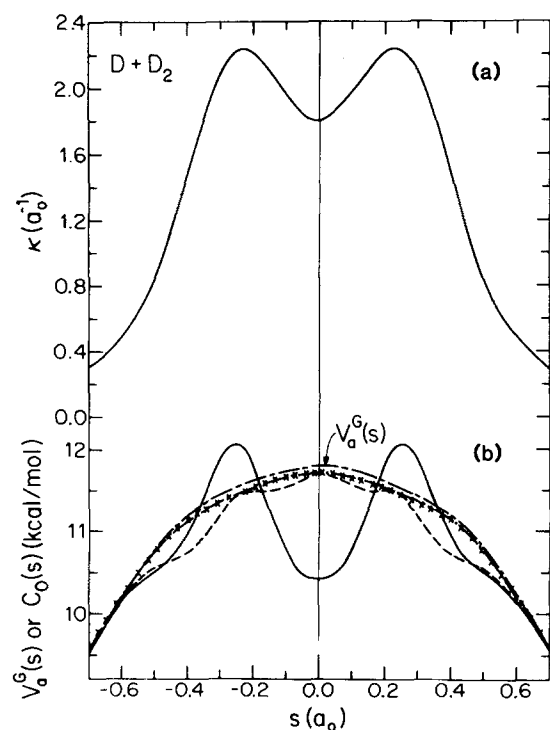


FIG. 2. Various quantities as functions of reaction coordinate s for the collinear $D+D_2$ reaction. (a) The solid curve is the curvature $\kappa(s)$ of the minimum-energy path. (b) $V_a^G(s)$ is shown as ---, and $C_0(s)$ is shown as solid curve. Other curves are $C_0(s)$ with various arguments set equal to zero: ---, $\kappa''=0$; xxxxx, $\kappa''=\kappa'=0$; -.-.-, $\kappa''=\kappa'=\alpha''=\alpha'=0$.

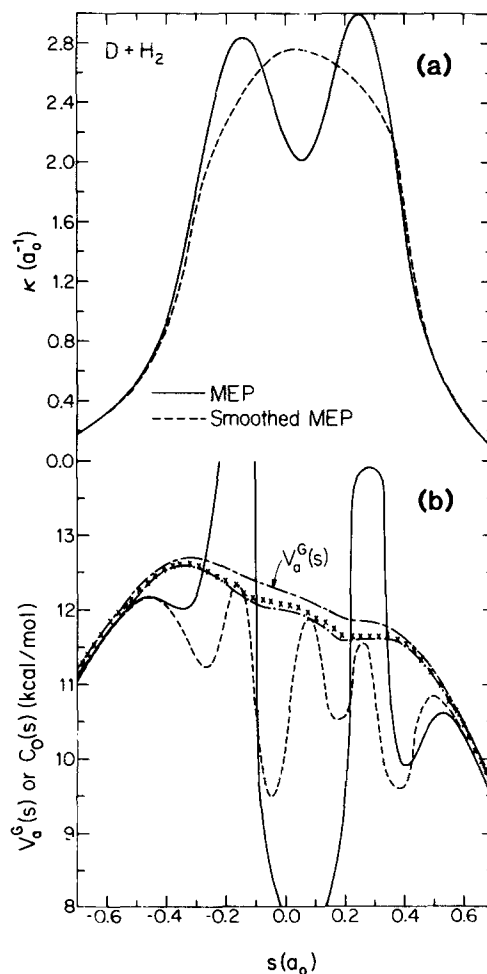


FIG. 3. Same as Fig. 2 except for $D+H_2$ and the dashed curve in part a is the curvature of the smoothed reaction path of Fig. 1. The solid curve in part a is the curvature of the minimum-energy path of Fig. 1.

dimensional ground-state Hamiltonian of Eq. (20) by

$$T(E) = \{1 + \exp[2\theta(E)]\}^{-1}, \quad (29)$$

where

$$\theta(E) = \frac{1}{\hbar} \int_{s_2}^{s_1} \left(-B_0(s) + \left\{ B_0^2(s) + \frac{2}{\mu_{\text{eff}}^{\text{PA}}(s, 0)} [C_0(s) - E] \right\}^{1/2} \right) \mu_{\text{eff}}^{\text{PA}}(s, 0) ds, \quad (30)$$

and where s_2 are the classical turning points defined by

$$C_0(s_2) = E. \quad (31)$$

In deriving Eq. (30), the fact that $B_0(s)$, $C_0(s)$, and $\mu_{\text{eff}}^{\text{PA}}(s, 0)$ depend on $\hbar$ was ignored; otherwise, the $\hbar$ -dependent parts would only affect the semiclassical transmission probability in a higher order approximation. Figures 4 and 5 show that the nonadiabatic derivative terms have a major effect on the reaction probability. Again the κ' and κ'' terms are the most important. Figures 2(a) and 3(a) show plots of $\kappa(s)$ for the two systems. By comparing Figs. 2(a) and 3(a) to Figs. 2(b) and 3(b), the relationship between the potential $C_0(s)$ and $\kappa(s)$ is easily seen.

The effects of the κ' and κ'' nonadiabatic-derivative terms are very large, and we will argue that they are unphysical. The origin of these terms is the definition of the MEP. Consider, hypothetically, a slightly dis-

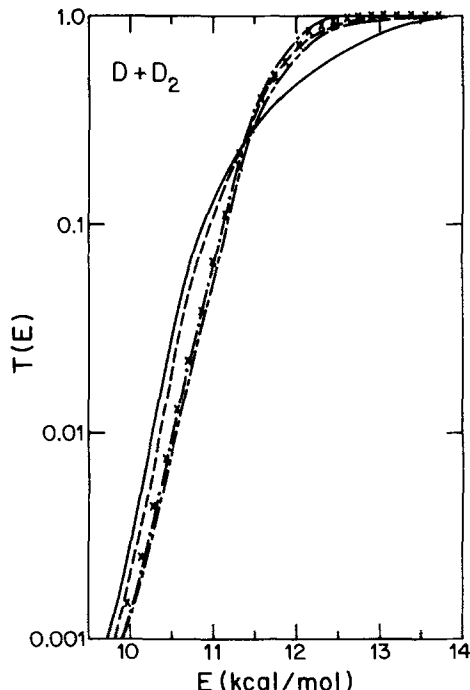


FIG. 4. Transmission probabilities for collinear $D + D_2$ as functions of total energy E computed by Eqs. (29) and (30) with $A_0(s)$, $B_0(s)$, and $C_0(s)$ from Eqs. (25)–(27). The solid curve is the full result. The other curves correspond to setting various arguments to zero in $B_0(s)$ and/or $C_0(s)$: ---, $\kappa'' = 0$; xxxxx, $\kappa'' = \kappa' = 0$; -.-.-, $\kappa'' = \kappa' = \alpha'' = \alpha' = 0$; ---, $\kappa'' = \kappa' = \alpha'' = \alpha' = \kappa = 0$. Notice that the --- curve corresponds to the harmonic PASAG result for which $C_0(s)$ is replaced by $V_a^G(s)$.

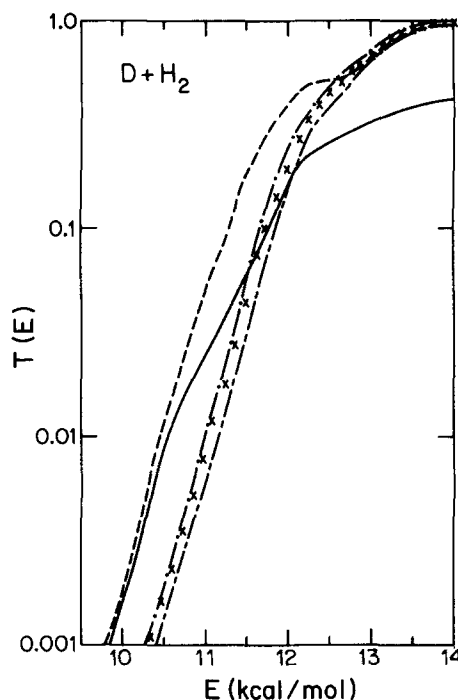


FIG. 5. Same as Fig. 4 except for $D + H_2$.

placed reaction coordinate such as the one shown by the dashed curve in Fig. 1 for $D + H_2$. This path smooths the curvature, as shown in Fig. 3(a), thus greatly reducing the magnitude of κ' and κ'' (see Fig. 3) and of the nonadiabatic derivative terms, while it displaces the reaction coordinate from the MEP by at most about $0.01a_0$. Furthermore, the potential quantities such as $V_a^G(s)$ and $\omega(s)$ are altered by only 0.1 kcal/mol or less by the change of coordinates. So a minor coordinate alteration which yields a small shift in the adiabatic potential curve gives large effects on the nonadiabatic derivative terms. It seems unlikely that the system would be so maximally adiabatic in a particular coordinate system that the highly unstable (with respect to coordinate variations) κ' and κ'' terms could be taken seriously. Instead, we suspect that there will be only a weak response to the slight coordinate variations aforementioned. Thus, our procedure will be to ignore the κ' and κ'' terms for one-dimensional calculations. Our presumption is that if a many-basis-function calculation were carried out including these terms, then the effects of the nonadiabatic derivative terms would tend to cancel out, leading to final results more in keeping with a one-dimensional calculation without them.

The remaining nonadiabatic derivative terms contain α' and α'' . These terms represent the diagonal part of the nonadiabatic derivative coupling arising from the narrowing and widening of the vibrational well. We see from Figs. 2 and 3 that the effect of these terms on relevant physical quantities is small. Thus, although there is no clear cut physical rationale for discarding these terms as there was for the κ' and κ'' terms, we nevertheless choose to drop them. We do so for two reasons. First, since the terms clearly arise from the

nonadiabatic coupling, they are more relevant to many-basis-functions calculations than to one-dimensional models. Second, the numerical magnitude of the effects of these terms is small compared to differences due to alternative methods of performing the expectation values, making adiabatic separations, and choosing the coordinate system. Furthermore, the effects of these terms lead to smaller differences in calculated reaction probabilities than the differences between any of the approximate methods considered and the exact results. Thus, we choose to neglect all nonadiabatic derivative terms for one-dimensional models.

Although we suppose that the κ' and κ'' terms do not have large effects, there is much less freedom with regard to the $\kappa(s)$ terms. This is primarily due to the constraint²²

$$\int_{-\infty}^{\infty} \kappa(s) ds = \pi - \beta, \quad (32)$$

where β is the skew angle⁹ given by

$$\beta = \arctan [m_B(m_A + m_B + m_C)/m_A m_C]^{1/2}. \quad (33)$$

Thus when $\kappa(s)$ is modified it must satisfy Eq. (32); a simple smoothing such as that shown in Fig. 3(a) changes the value of $\kappa(s)$ at various values of s , but not drastically, although such a smoothing does reduce κ' and κ'' considerably.

The one final troublesome term in Eq. (20) is the curvature zero point term

$$V^Z(s) \equiv -\frac{\hbar^2 \kappa^2(s)}{8\mu_{\text{eff}}(s, 0)}. \quad (34)$$

In many cases this term would make only a minor contribution to the transmission probabilities. In other cases the effect would be larger. But this term has an explicit dependence on $\hbar$, and it is not clear to what order it should enter into semiclassical calculations. In the rest of this paper we neglect it.

From the considerations presented above, the defining equation may be written in the following way for a general state n [compare Eq. (20)]:

$$\left[-\frac{\hbar^2}{2\mu} A_n(t) \frac{\partial^2}{\partial t^2} + V_{\text{ave}}(t, n) - E \right] f_n(t) = 0, \quad (35)$$

where

$$A_n(t) = \int_{-\infty}^{1/\kappa(t)} |\chi_n(v; t)|^2 \frac{1}{\eta^2} dv, \quad (36)$$

$$V_{\text{ave}}(t, n) = \int_{-\infty}^{1/\kappa(t)} |\chi_n(v, t)|^2 [V_1(v, t) + \epsilon_n(t)] dv. \quad (37)$$

Two of the methods (SO and SOP) to be considered in Sec. V also include the effects of virtual couplings to the other adiabatic states $\chi_m(v, t)$ for $m \neq n$; the contribution of these couplings to the effective potential will be called $V_{\text{pert}}(t, n)$. Thus,

$$\left[-\frac{\hbar^2}{2\mu_{\text{eff}}(t, n)} \frac{\partial^2}{\partial t^2} + V_{\text{ave}}(t, n) + V_{\text{pert}}(t, n) - E \right] f_n(t) = 0, \quad (38)$$

where we have defined the effective reduced mass as

$$\mu_{\text{eff}}(t, n) \equiv \frac{\mu}{A_n(t)}. \quad (39)$$

Equation (38) is the general equation from which all subsequent models considered in this article are developed. There are three items of choice in further specifying a one-dimensional model on the basis of this equation: (1) choosing a coordinate system; (2) making an adiabatic separation of the Hamiltonian, i.e., identifying $V_2(v; t)$ and $V_{\text{pert}}(t, n)$; and (3) choosing a method for performing the expectation values. After these choices are made, Eqs. (16), (19b), (36), (37), and (39) automatically yield the substitutions for the general equation.

The curvature explicitly enters Eq. (35) through the quantity $A_n(t)$, which we prefer to recast physically as the effective reduced mass $\mu_{\text{eff}}(t, n)$. It is in the calculation of $\mu_{\text{eff}}(t, n)$ that we discover the problems of doing the expectation values. A purely formal definition of the effective reduced mass is

$$\frac{1}{\mu_{\text{eff}}(t, n)} = \frac{1}{\mu} \int_{-\infty}^{1/\kappa(t)} \frac{|\chi_n(v; t)|^2}{[1 - \kappa(t)v]^2} dv. \quad (40)$$

We see that unless the limit

$$L(\epsilon) = \lim_{v \rightarrow 1/\kappa(t)} \left\{ \frac{|\chi_n(v; t)|^2}{[1 - \kappa(t)v]^{1+\epsilon}} \right\} \quad (41)$$

is finite for some $\epsilon > 0$, the integral diverges. The only way for this to happen for all n is to include in the vibrational Schrödinger equation for $\chi_n(v; t)$ a potential term with a repulsive pole at $v = 1/\kappa(t)$. This constrains $\chi_n(v; t)$, $\phi_n(v; t)$ of Eq. (13) and the total wave function $\psi(t, v)$ of Eq. (9) all to be zero at $[t, 1/\kappa(t)]$. Not only is it physically unreasonable to compel the wave function to be zero at $[t, 1/\kappa(t)]$, but one can get any value for $A_n(t)$ by varying the rate at which it goes to zero there, i.e., by choosing different singular potentials. To make matters even worse, the changes in $\mu_{\text{eff}}(t, n)$ and in $V_{\text{ave}}(t, n)$ with variations in the singular potential do not even approximately cancel each other in the semiclassical momentum which is discussed in the next section. These problems are pathologies of the coordinate system related to the existence of the multivalued region. Since there is no physical basis on which to identify a singular potential which forces the wave function to an unphysical zero, we feel there is no choice but to abandon the formal definition of the expectation value in favor of a physically reasonable interpretation.²³ In Sec. V, a variety of models for $\mu_{\text{eff}}(t, n)$ are discussed.

IV. SEMICLASSICAL TRANSMISSION COEFFICIENTS

The WKB method is used to solve the differential Eq. (38). This method is chosen as the best compromise between accuracy and simplicity. We consider transmission across or through the ground-state potential only. Accordingly, for energies below the barrier maximum the reaction probability is given by Eq. (29) with

$$\theta(E) = \frac{1}{\hbar} \int_{t_c}^{t_s} \text{Im } p(t, E) dt. \quad (42)$$

The imaginary momentum $\text{Im } p(t, E)$ is found semiclassically by replacing $\hbar/i \partial/\partial t$ by $p(t, E)$ in Eq. (38). The resulting expression is

$$p(t, E) = (2\mu_{\text{eff}}(t, 0) \{E - V_{\text{ave}}^G(t) - V_{\text{pert}}^G(t)\})^{1/2}, \quad (43a)$$

where

$$V_{\text{ave}}^G(t) = V_{\text{ave}}(t, n=0) \quad (43b)$$

and

$$V_{\text{pert}}^G(t) = V_{\text{pert}}(t, n=0). \quad (43c)$$

When $V_{\text{ave}}^G(t)$ and $V_{\text{pert}}^G(t)$ are independent of energy, Eq. (42) becomes

$$\theta(E) = \frac{1}{\hbar} \int_{t_c}^{t^*} \{2\mu_{\text{eff}}(t, 0) [V_{\text{ave}}^G(t) + V_{\text{pert}}^G(t) - E]\}^{1/2} dt, \quad E \leq V_p^G, \quad (44a)$$

where

$$V_p^G \equiv \max_t [V_{\text{ave}}^G(t) + V_{\text{pert}}^G(t)]. \quad (44b)$$

For energies above V_p^G we use a parabolic approximation for nonclassical reflection.²¹ In this approximation we still use Eq. (29), but the phase integrals for energies above V_p^G are given by

$$\theta(V_p^G + \Delta) = -\theta(V_p^G - \Delta), \quad 0 < \Delta \leq V_p^G - \epsilon_0^{\text{max}} \quad (45a)$$

and

$$T(E) = 1, \quad E > 2V_p^G - \epsilon_0^{\text{max}}, \quad (45b)$$

where

$$\epsilon_0^{\text{max}} = \max(\epsilon_0^R, \epsilon_0^P) \quad (45c)$$

and

$$\epsilon_n^R = V_{\text{ave}}(t = -\infty, n) + V_{\text{pert}}(t = -\infty, n), \quad (45d)$$

$$\epsilon_n^P = V_{\text{ave}}(t = +\infty, n) + V_{\text{pert}}(t = +\infty, n). \quad (45e)$$

This is called the semiclassical adiabatic ground-state (SAG) method. For some of the tunneling approximations (viz., the MCP and SC methods) considered below, the physical considerations determining $\mu_{\text{eff}}(t, 0)$ in the tunneling regime are not appropriate for the classical-reflection problem. In such cases it may be better to use a different method to calculate $\theta(V_p^G - \Delta)$ for Eq. (45a) than is used for $\theta(E)$ in Eq. (29) at tunneling energies. Two methods, the minimum-energy-path (MEP) method and the dynamical-path vibrational-average (DA) method, will be considered. In the former we use

$$\mu_{\text{eff}}^{\text{MEP}}(s, 0) = \mu, \quad (46a)$$

$$V_{\text{ave}}^{\text{MEP}}(s, 0) = V_a^G(s), \quad (46b)$$

$$V_{\text{pert}}^{\text{MEP}}(s, 0) = 0, \quad (46c)$$

and the phase integral is

$$\theta^{\text{MEP}}(V_p^G + \Delta) = \hbar^{-1} \int_{s_c}^{s^*} ds \{2\mu[-V_a^G(s) + V_p^G + \Delta]\}^{1/2}. \quad (47)$$

This corresponds to including reaction-path curvature only in the tunneling region. In general we put a prefix (e.g., PA, MCP, SC, or DA) before SAG to denote the method used to include reaction-path curvature in the SAG transmission coefficient. If we include reaction-path curvature only at tunneling energies we add a suffix *T* to the prefix, e.g., SCSAG (small-curvature-approximation SAG) denotes using Eqs. (29), (43a), (44a), (45a), and (45b) with the SC approximations for

the quantities in Eq. (43a), but SCTSAG (small-curvature-tunneling SAG) denotes using Eq. (47) for $\theta(V_p^G + \Delta)$ in Eq. (45a). Another possibility is to compute nonclassical reflection by the dynamical-path vibrational-average method of Sec. VC. This will be denoted by appending da to the prefix, e.g., SCdaSAG. In summary: (i) if we use the small-curvature (SC) approximation for $\theta(E)$ at tunneling energies and for $\theta(V_p^G + \Delta)$ for calculating nonclassical reflection, we call the result SCSAG; (ii) if we use the SC approximation for $\theta(E)$ at tunneling energies but the MEP approximation for $\theta(V_p^G + \Delta)$, we call the result SCTSAG; and (iii) if we use SC approximation for $\theta(E)$ at tunneling energies but the DA approximation for $\theta(V_p^G + \Delta)$, we call the result SCdaSAG.

In all cases, the SAG transmission coefficients are computed as the ratio of the thermally averaged semiclassical reaction probability to the thermally averaged classical one,^{13,21} i.e.,

$$\kappa(T) = \langle T_C \rangle^{-1} \int_{\epsilon_0^{\text{max}}}^{\infty} T(E) \exp(-\beta E) dE, \quad (48a)$$

where

$$\langle T_C \rangle = \int_{V_{\text{AG}}}^{\infty} \exp(-\beta E) dE \quad (48b)$$

and

$$V_{\text{AG}} = \max_t V_a^G(t). \quad (48c)$$

The expression in Eq. (48b) is the thermally averaged ground-state reaction probability for classical reaction-coordinate motion with no reaction-path curvature.

We anticipate the results to be obtained below and note here that in three cases (DA, SO, and SOP) that we will consider for which $V_{\text{ave}}^G(t) + V_{\text{pert}}^G(t)$ does not equal the zero-curvature $V_a^G(t)$, we will still have V_p^G equal to V_{AG} . Hence we will have tunneling for $E < V_{\text{AG}}$ and nonclassical reflection for $E > V_{\text{AG}}$. Furthermore, we will see that $\theta(E = V_{\text{AG}})$ is zero in all these methods so that $T(E)$ is continuous even if different methods are used for tunneling and for nonclassical reflection.

V. ADIABATIC MODELS

In this section we consider only the ground state of collinear A + BC reactions; furthermore in Secs. A–G only the harmonic approximation for the vibrational motion is considered. Extensions to higher *n* are possible in all cases but to present them here would complicate the formulas with little gain in insight. The anharmonic extensions, if known, are presented in Sec. H. As mentioned at the end of Sec. II, care has been taken to distinguish the coordinate system used in each method: (*t, v*) denotes the generic coordinate system, (*s, u*) specifies the MEP coordinates, and (*ρ, ω*) specifies the DP coordinates.

A. Phase-average approximation

The first one-dimensional model, the classical phase average (PA) method, has already been considered briefly in Sec. III and in more detail in Refs. 11 and 13. The PA method was originally proposed by Miller *et al.*¹¹; it was extended to include anharmonicity and

tested in a previous paper by our group.¹³ With the PA method one adopts the MEP coordinate system and makes the simplest adiabatic separation of the potential, viz.

$$V_1(s, u) = V(s, u=0) \equiv V^{\text{MEP}}(s), \quad (49a)$$

$$V_2(u; s) = V(s, u) - V^{\text{MEP}}(s), \quad (49b)$$

$$V_{\text{pert}}(s, n) = 0. \quad (49c)$$

At the heart of the PA method is an intuitive physical picture of vibrational motion with the following two properties. First, at any s the vibration acts like a simple one-dimensional bound motion in the well described by $V_2(u; s)$ and is free from any centrifugal distortions arising from s motion. And second, the necessary expectation values can be performed using classical phase averaging. Consequently, in the case of small curvature, as described by Eq. (7), the vibrational coordinate is never distended into the multivalued region and $A_n(s)$ will be finite, thus making $\mu_{\text{eff}}(s, n)$ well defined. According to this view, it is the amplitude near the minimum of the potential well rather than that near the multivalued region which mainly determines the behavior of the system. The classical phase average, which cuts off the dangerous tail of the wave function because it includes no contributions from beyond the turning point, enforces this perspective in the computation of the effective mass.

The classical formalism of Miller *et al.*¹¹ is now briefly reviewed because it serves as an introduction to the methods considered in later subsections. For a collinear system the classical Hamiltonian with harmonic vibrational motion is

$$H_C(p_s, s, p_u, u) = \frac{p_s^2}{2\mu[1 - \kappa(s)u]^2} + \frac{p_u^2}{2\mu} + V^{\text{MEP}}(s) + \frac{1}{2}\mu\omega^2(s)u^2. \quad (50)$$

Expressed in action-angle variables,²⁴ $[(n + \frac{1}{2})\hbar, q]$, for the u motion, Eq. (50) becomes

$$H_C(p_s, s, n, q) = \frac{1}{2\mu} \left\{ \frac{p_s + (n + \frac{1}{2})\hbar \left[\frac{\omega'}{\omega(s)} \right] \cos q \sin q}{1 + \left[(2n + 1) \frac{\hbar \kappa^2(s)}{\mu \omega(s)} \right]^{1/2} \sin q} \right\}^2 + (n + \frac{1}{2})\hbar \omega(s) + V^{\text{MEP}}(s). \quad (51)$$

Since the classical expression of adiabaticity is conservation of the action, the one-dimensional effective Hamiltonian for $n=0$ is the angle-averaged expression

$$H_C^a(p_s, s) = \frac{1}{2\pi} \int_0^{2\pi} H_C(p_s, s, n=0, q) dq, \quad (52)$$

which yields

$$H_C^a(p_s, s) = \frac{p_s^2}{2\mu[1 - \gamma^2(s)]^{3/2}} + \frac{\hbar^2}{8\mu} \left[\frac{\omega'}{\omega(s)} \right]^2 f(\gamma) + V^{\text{MEP}}(s) + \epsilon_0(s), \quad (53)$$

where

$$f(\gamma) = \left\{ \frac{\gamma^2(s)}{2} - 3 + \frac{3 - 2\gamma^2(s)}{[1 - \gamma^2(s)]} \right\} \gamma^4(s), \quad (54)$$

$$\gamma(s) = \left[\frac{\hbar}{\mu \omega(s)} \right]^{1/2} \kappa(s) = \alpha(s). \quad (55)$$

The ω' term is generally small and was discarded in Ref. 13 where the resulting adiabatic model was called the phase-average (PA) model. In the PA model, Eq. (53) can be used to specify the remaining quantities in the general equation:

$$\mu_{\text{eff}}^{\text{PA}}(s, 0) = \mu[1 - \gamma^2(s)]^{3/2} \quad (56)$$

and

$$V_{\text{ave}}^{\text{PA}}(s, 0) = V_a^G(s), \quad (57)$$

$$V_{\text{pert}}(s, 0) = 0. \quad (58)$$

Re-introduction of the ω' terms yields a new model which could be called PA ω' for consistency with the SO ω' model discussed below.

B. Vibrational-average method

A method closely related in spirit to the phase-average method is the quantum vibrational-average (VA) model. The MEP coordinate system is chosen, and the same adiabatic separation as in the PA method is used. The difference between the VA and PA methods is that in the VA model a quantal expectation value is used to compute the effective reduced mass. We again assume no centrifugal effect on the vibrational motion and that the dominant effects of curvature take place near the minimum of the vibrational well. However, instead of using a classical model to perform expectation values an alternative procedure is used. To avoid the divergence in $A_n(s)$ due to the singularity of η^{-2} , a Taylor expansion of η^{-2} around $u=0$ is used:

$$\eta^{-2} \approx 1 + 2\kappa(s)u + 3\kappa^2(s)u^2 \quad (59)$$

and so

$$A_0^{\text{VA}}(s) = \int_{-\infty}^{\infty} [1 + 2\kappa(s)u + 3\kappa^2(s)u^2] |\chi_0(u; s)|^2 du \quad (60)$$

and

$$\mu_{\text{eff}}^{\text{VA}}(s, 0) = \mu/A_0^{\text{VA}}(s). \quad (61)$$

The purpose of this procedure is to accurately represent η^{-2} near $u=0$ where the dominant effects are assumed to occur while avoiding the singularity at $u=1/\kappa(s)$ due to the remainder of the Taylor series. When $\mu_{\text{eff}}^{\text{VA}}(s, 0)$ is evaluated in the harmonic approximation we get

$$\mu_{\text{eff}}^{\text{VA}}(s, 0) = \mu/[1 + \frac{3}{2}\gamma^2(s)]. \quad (62)$$

Otherwise the VA and PA methods are identical.

C. Dynamical-path vibrational-average method

Use of the dynamical path (DP) allows for a more realistic inclusion of the effects of curvature of the reaction coordinate. The goal of the method is to include the energy-dependent centrifugal distortion of the vibrational wave function as well as the effective-reduced-mass correction to the reaction-coordinate motion. The hope is that in this way a more accurate adiabatic separation of the Hamiltonian can be achieved.

The separation of the Hamiltonian we present here was originally derived by Marcus¹⁵ using a separation-of-variables technique; Marcus' formulas were subsequently applied to $\text{H} + \text{H}_2$ by Wyatt¹⁶ and by Bowman.¹⁹ We shall rederive the Marcus result using a simpler dynamical argument and cast the result in the form of the general Eq. (38).

We represent the curvature effects entering into the vibrational equation of motion as a centrifugal potential. The vibrational problem including the centrifugal potential will be called the distorted vibrational problem, and the quantities involved in it will be denoted by tildes. The centrifugal potential can be obtained by introducing various approximations in the full Schrödinger Eq. (10b). Using a ground-state adiabatic wave function of the form

$$\psi(t, v) = \eta^{-1/2} f_0(t) \tilde{\chi}_0(v; t) \quad (63)$$

in Eq. (10b), neglecting curvature zero-point and non-adiabatic derivative terms, and replacing the operator $\hbar/i \partial/\partial t$ by the local reaction-coordinate momentum $p(t, E)$, which is assumed to be independent of v , we obtain

$$\frac{f_0(t)}{\eta^{1/2}} \left\{ -\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial v^2} + \frac{[p(t, E)]^2}{2\mu \eta^2} + V(v, t) - E \right\} \tilde{\chi}_0(v, t) = 0. \quad (64)$$

Define $v_0(t)$ as the position of the minimum of the effective potential:

$$\frac{\partial}{\partial v} \left\{ V(v, t) + \frac{[p(t, E)]^2}{2\mu \eta^2} \right\} \bigg|_{v=v_0(t)} = 0 \quad (65)$$

and define the dynamical path such that $v = v_0(t)$ along it. The numerical value for $p(t, E)$ is obtained by writing the total energy at (t, v_0) as the sum of the potential energy, the zero point energy $\tilde{\epsilon}_0(t)$ of the distorted vibrational problem, and the reaction-coordinate translational energy, i. e.,

$$E = V(t, v_0) + \tilde{\epsilon}_0(t) + \frac{[p(t, E)]^2}{2\mu \eta_0^2(t)}, \quad (66)$$

where

$$\eta_0(t) = 1 - \kappa(t) v_0(t). \quad (67)$$

Combining Eqs. (64) and (66) yields

$$\left\{ -\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial v^2} + \left[\frac{p^2(t, E)}{2\mu \eta^2} + V(t, v) - \frac{p^2(t, E)}{2\mu \eta_0^2(t)} - V(t, v_0) \right] - \tilde{\epsilon}_0(t) \right\} \tilde{\chi}_0(v; t) = 0. \quad (68)$$

Comparing Eqs. (14) and (68), remembering that the curvature zero point term [Eq. (34)] is being neglected, and using Eq. (66) yields the identification

$$V_2(v; t) = V(t, v) - V(t, v_0) + \left[\frac{\eta_0^2(t)}{\eta^2} - 1 \right] [E - \tilde{\epsilon}_0(t) - V(t, v_0)]. \quad (69)$$

In this formalism one must solve iteratively and self-consistently for $v_0(t)$ and $\tilde{\epsilon}_0(t)$ at each t . In the harmonic approximation below however we can avoid iteration.

Given the vibrational potential $V_2(v; t)$ and taking $V_{\text{pert}}(t, n) = 0$, the distorted versions of Eqs. (36), (37),

and (38) yield the one-dimensional wave equation

$$\left\{ -\frac{\hbar^2}{2\mu} \tilde{A}_0(t) \frac{d^2}{dt^2} + V(t, v_0) - [\eta_0^2(t) \tilde{A}_0(t) - 1] \right\} \times [E - \tilde{\epsilon}_0 - V(t, v_0)] + \tilde{\epsilon}_0(t) - E \Big\} f_0(t) = 0. \quad (70)$$

Using Eq. (66) reduces this to

$$\left\{ -\frac{\hbar^2}{2\mu \eta_0^2(t)} \frac{d^2}{dt^2} + V(t, v_0) + \tilde{\epsilon}_0(t) - E \right\} f_0(t) = 0, \quad (71)$$

where $\tilde{\epsilon}_0(t)$ is the eigenvalue in

$$\left\{ -\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial v^2} + V_2(v; t) - \tilde{\epsilon}_0(t) \right\} \tilde{\chi}_0(v; t) = 0. \quad (72)$$

Equations (71) and (72) are identical to Marcus' result¹⁵ with the exception of the curvature zero point term which he retained. It should be pointed out that the expectation value $\tilde{A}_0(t)$ completely cancels out of Eq. (71). The DA method is unique among the models we consider in not requiring the evaluation of this ambiguous quantity. In place of $\tilde{A}_0(t)$, the quantity $[\eta_0(t)]^{-2}$ determines the effective reduced mass. We shall see that this term has a geometrical interpretation analogous to that considered below for the Marcus-Coltrin-path and small-curvature-approximation models.

So far the choice of coordinate systems has been left unspecified. Inspection of Eq. (71) suggests that a convenient choice would be a coordinate system where $v_0 = 0$ (i. e., $\eta_0 = 1$). These are the DP coordinates (ρ, ω) . When written in DP coordinates, the imaginary momentum of WKB theory is simply

$$\text{Im } p(\rho, E) = \{2\mu [V(\rho, 0) + \tilde{\epsilon}_0(\rho) - E]\}^{1/2} \quad (73)$$

and the tunneling integral is

$$\theta(E) = \frac{1}{\hbar} \int_{\rho_L}^{\rho_R} \text{Im } p(\rho, E) d\rho. \quad (74)$$

Further, we can now recognize the factor $1/\eta_0^2(\rho)$ as a metric quantity which relates length along the MEP to ρ , the arc length of the DP tunneling path. In the tunneling region the negative centrifugal force pushes the dynamical path to the concave side of the MEP thereby shortening the tunneling path and increasing the transmission probability.

A difficulty with the dynamical-path model was pointed out by Wyatt.¹⁶ He found, using MEP coordinates, that for sufficiently negative local kinetic energies the vibrational potential does not have a local maximum between the minimum-energy path and the center of curvature. In a sense, the particle falls into a potential hole at the center of curvature. Consequently, the solution procedure for $v_0(t)$ and $\tilde{\epsilon}(t)$ fails immediately. This problem can be circumvented by the choice of a different coordinate system for the zeroth-order computation. By choosing a reaction coordinate sufficiently displaced to the concave side of the MEP, $\kappa(t)$ is lowered, the centrifugal force lessened, and the zeroth-order result becomes finite. This will not be required for the calculations in the present paper, but in principle it extends the range of practical applications of the dynamical-path approach.

We will now develop an approximation scheme in

which much of the calculation can be done analytically and noniteratively. Three separate approximations are made. (i) The Born–Oppenheimer potential is assumed to be harmonic about the MEP. (ii) The curvature is assumed small in the sense that the parameter $a(s)$ of Eq. (8) is much less than 1. (iii) It is assumed that the local kinetic energy is small enough so that the dynamical path is displaced from the minimum-energy path by only a small distance. We begin by rewriting Eq. (69) in the harmonic approximation in MEP coordinates

$$V_2(u; s) = V(s, u) - V(s, u_0) + \left[\frac{\eta_0^2(s)}{\eta^2} - 1 \right] [E - \tilde{\epsilon}_0(s) - V(s, u_0)] \\ = \frac{1}{2} \mu \omega^2(s) u^2 - \frac{1}{2} \mu \omega^2(s) u_0^2(s) + \left[\frac{\eta_0^2(s)}{\eta^2} - 1 \right] [E - \tilde{\epsilon}_0(s) \\ - V^{\text{MEP}}(s) - \frac{1}{2} \mu \omega^2(s) u_0^2(s)] . \quad (75)$$

The minimum of $V_2(u, s)$ is found by the condition

$$\left. \frac{\partial V_2(u; s)}{\partial u} \right|_{u=u_0(s)} = 0 \quad (76)$$

to be

$$u_0(s) = \frac{1 - [1 + 16a^2(s)\tilde{b}(s)]^{1/2}}{4\kappa(s)} , \quad (77)$$

with

$$\tilde{b}(s) = T(s)/\hbar\omega(s) \quad (78)$$

and

$$T(s) = E - \tilde{\epsilon}_0(s) - V^{\text{MEP}}(s) . \quad (79)$$

The quantity $\tilde{b}(s)$ is a dimensionless parameter equal approximately to the ratio of the local kinetic energy to the harmonic spacing of the undistorted vibration. [The actual local kinetic energy along the reaction coordinate is given by $E - \tilde{\epsilon}_0(s) - V^{\text{DP}}(s)$ instead of Eq. (79), where $V^{\text{DP}}(s)$ is $V(s, u_0)$.] Assumption (iii) is thus $\tilde{b}(s) \ll 1$. To second order in $a(s)$ and $\tilde{b}(s)$, $u_0(s)$ is

$$u_0(s) \approx -2 \frac{a^2(s)\tilde{b}(s)}{\kappa(s)} + \frac{8\tilde{b}^2(s)a^4(s)}{\kappa(s)} . \quad (80)$$

The potential $V_2(u; s)$ supports a harmonic bound state with a frequency $\tilde{\omega}(s)$ and harmonic energy levels $\tilde{\epsilon}_0(s)$ given by

$$\left. \frac{\partial^2 V_2(u; s)}{\partial u^2} \right|_{u=u_0(s)} = \mu \tilde{\omega}^2(s) \quad (81)$$

and

$$\tilde{\epsilon}_0(s) = \frac{1}{2} \hbar \tilde{\omega}(s) . \quad (82)$$

Analogous to $\tilde{b}(s)$ we have a similar quantity $b(s)$ for the undistorted vibration

$$b(s) = \frac{E - \epsilon_0(s) - V^{\text{MEP}}(s)}{\hbar\omega(s)} . \quad (83)$$

Using the undistorted quantities $b(s)$ and $\omega(s)$ we wish to obtain a small- $a(s)$ and small- $b(s)$ approximation to

$$\Delta\omega(s) = \tilde{\omega}(s) - \omega(s) . \quad (84)$$

Using Eqs. (75) and (78)–(80) we obtain

$$\frac{\tilde{\omega}^2(s)}{\omega^2(s)} = 1 + \frac{6a^2(s)\tilde{b}(s)}{[1 + 2a^2(s)\tilde{b}(s) - 8a^4(s)\tilde{b}^2(s)]^2} \\ \times \left\{ 1 - \frac{1}{2a^2(s)\tilde{b}(s)} [2a^2(s)\tilde{b}(s) - 8a^4(s)\tilde{b}^2(s)]^2 \right\} . \quad (85)$$

To low order in $\tilde{b}(s)$ and $a(s)$, Eq. (85) reads

$$\frac{\tilde{\omega}^2(s)}{\omega^2(s)} = 1 + 6a^2(s)\tilde{b}(s) - 36a^4(s)\tilde{b}^2(s) . \quad (86)$$

Using Eq. (81) and

$$\tilde{b}(s) = b(s) - \frac{\Delta\omega(s)}{2\omega(s)} , \quad (87)$$

Eq. (86) can be written as a quadratic in $\Delta\omega(s)$, whose solution to low order in $a(s)$ and $b(s)$ is

$$\Delta\omega(s) \approx 3a^2(s)\omega(s)b(s) + 0[a^4(s)b(s)] . \quad (88)$$

Equations (80), (87), and (88) obviate the need for iteration.

We complete the calculation by finding an expression for the semiclassical imaginary momentum. Since the MEP and DP reaction paths are close to one another when the assumption of small $\tilde{b}(s)$ holds, we do the calculation in MEP coordinates. When $\tilde{b}(s)$ is small, the metric factor relating the s and ρ arc lengths is approximately $1/\eta_0(s)$; in other words we have the following relations:

$$ds \approx \eta_0^{-1}(s) d\rho , \quad (89a)$$

$$\text{Im } p(s, E) = \text{Im } p(\rho, E) \eta_0(s) . \quad (89b)$$

Using Eqs. (73), (77)–(79), (89), and

$$V(s, u) = V^{\text{MEP}}(s) + \frac{1}{2} \mu \omega^2(s) u^2 \quad (90)$$

yields

$$\theta(E) = \int_{s_<}^{s_>} ds \text{Im } p(s, E) , \quad (91)$$

where

$$\text{Im } p(s, E) = \left\{ -2\mu\eta_0^2(s) [\tilde{b}(s)\hbar\omega(s) - \frac{1}{2}\mu\omega^2(s)u_0^2(s)] \right\}^{1/2} . \quad (92)$$

To low order in $a(s)$

$$\tilde{b}(s) \approx b(s) \mu_{\text{eff}}^{\text{DA}}(s, 0) / \mu , \quad (93)$$

where

$$\mu_{\text{eff}}^{\text{DA}}(s, 0) = \mu \left[1 - \frac{3}{2} a^2(s) \right] . \quad (94)$$

This yields

$\text{Im } p(s, E)$

$$\approx \left\{ -2\mu_{\text{eff}}^{\text{DA}}(s, 0)b(s)\hbar\omega(s) \left[1 + \frac{2\mu_{\text{eff}}^{\text{DA}}(s, 0)a^2(s)b(s)}{\mu} \right] \right\}^{1/2} . \quad (95)$$

The radicand of Eq. (95) is more complicated than that of Eq. (44a); however, when $a(s)$ is small the complex roots of $\text{Im } p(s, E) = 0$ are located far from the real roots and the classical turning points are given by the solution of

$$b(s_*) = 0 . \quad (96)$$

At $E = V^{\text{AG}}$, the phase integral goes to zero and the transmission probability can be smoothly connected to the zero-curvature results [Eqs. (45a) and (46a)–(47)]

for energies above V_a^G . A second point of concern is the limiting behavior of Eq. (95). We see that for sufficiently low energies, $\text{Im } p(s, E)$ might maximize and then decline as the energy is lowered further. This unphysical effect is the result of the breakdown of assumption (iii). Since we wish to calculate thermally averaged transmission coefficients which require transmission probabilities at these low energies, and since the transmission coefficients are largely insensitive to the behavior at these low energies, we adopt the following *ad hoc* procedure to handle the problem. For energies below that at which the maximum of $\text{Im } p(s, E)$ is reached, for each s value, $\text{Im } p(s, E)$ is replaced by this maximum value for all lower energies. This yields

$$\text{Im } p(s, E) = \left[\frac{\mu \hbar \omega(s)}{4a^2(s)} \right]^{1/2}, \quad E < V_a^G(s) - \frac{\mu \hbar \omega(s)}{4\mu_{\text{eff}}^{\text{D,P}}(s, 0)a^2(s)}. \quad (97)$$

The low-order DA result (95), with the definitions (83) and (94), can be expressed using the general Eq. (43a) by making the following identifications:

$$V_{\text{ave}}^{\text{DA}}(t, 0) = V_a^G(s) - 2\hbar\omega(s)a^2(s)b^2(s)\mu_{\text{eff}}^{\text{DA}}(s, 0)/\mu, \quad (98)$$

$$V_{\text{pert}}^{\text{DA}}(t, 0) = 0. \quad (99)$$

In this way the effects of centrifugal distortion are included in the averaged potential computed for the dynamical path.

D. Marcus-Coltrin path method

The PA and VA methods make use of vibrational adiabaticity in two distinct ways: first in order to establish the adiabatic potential curve and second to interpret the expectation value $A_n(s)$. In the second use the vibrational wave function is assumed to maintain its form in the u coordinate even when the system is in a classically forbidden region. In the Marcus-Coltrin path (MCP) method this adiabatic interpretation of $A_n(s)$ is abandoned for the classically forbidden region. The MCP method is equivalent to assuming that the vibrational wave function collapses about a minimum-imaginary-action path wherever the system energy is less than the adiabatic potential curve or that the phase integral can be calculated along the caustic of concave-side turning points. The adiabatic potential curve is calculated in the same way as in the PA and VA methods, but the expectation values must be done in a different fashion. Since the MCP method has been thoroughly discussed elsewhere,^{4,7-9} we shall confine ourselves here to a reconciliation of the MCP method with the general Eq. (38).

In the MCP method we use the MEP coordinates and the potential separation of Eq. (49). The minimum-imaginary-action path is approximated within the framework of the following assumption. Marcus and Coltrin assumed that within the curved swath of $\{s, u\}$ space lying between the classical vibrational caustics, $u_<(s, n)$ and $u_>(s, n)$, the effective potential is $V_a^G(s)$. Beyond the caustics the effective potential is the Born-Oppenheimer potential. The minimum-imaginary-action path from among a sequence of parametrized paths between $s_>$ and $s_<$ was found numerically for ground-state $\text{H} + \text{H}_2$ to the path $u_>(s, n)$ of outer turning points. This path, which is shorter than the MEP, is the MCP. The tun-

neling integral $\theta(E)$ of Eq. (44a) becomes

$$\theta(E) = \frac{1}{\hbar} \int_{\xi_<}^{\xi_>} (2\mu\{V_a^G[s(\xi)] - E\})^{1/2} d\xi, \quad (100)$$

where ξ measures distance along the MCP. We can see how to obtain this result from the general Eq. (38) by comparing Eqs. (100) and (44a) for $\theta(E)$. This shows that $V_{\text{pert}}^G(t)$ is zero in the MCP method and that the effective reduced mass should be interpreted as a metric factor relating the differential lengths along the MCP and MEP.¹⁴ Thus,

$$d\xi = [\mu_{\text{eff}}^{\text{MCP}}(s, 0)/\mu]^{1/2} ds, \quad (101)$$

and from analytical geometry

$$\mu_{\text{eff}}^{\text{MCP}}(s, 0) = \mu\{[1 - \kappa(s)u_>(s)]^2 + (u_<'(s))^2\}. \quad (102)$$

Thus,

$$\text{Im } p(s, E) = \{2\mu_{\text{eff}}^{\text{MCP}}(s, 0)[V_a^G(s) - E]\}^{1/2}. \quad (103)$$

In the harmonic approximation the relevant turning point becomes

$$u_>(s) = \left[\frac{\hbar}{\mu\omega(s)} \right]^{1/2}, \quad (104)$$

so that $u_<'$ in Eq. (102) depends on ω' . Then the transmission probabilities can be calculated as before from Eqs. (29), (42), and (102)–(104) without having to explicitly calculate the MCP. In terms of the general Eq. (38) the MCP method is specified by using $\mu_{\text{eff}}^{\text{MCP}}(s, 0)$ of Eq. (102) and

$$V_{\text{ave}}^{\text{MCP}}(s, 0) = V_a^G(s), \quad (105)$$

$$V_{\text{pert}}^{\text{MCP}}(s, 0) = 0. \quad (106)$$

Notice that the MCP method as defined here is not identical to the MCP method as applied previously,^{7-9,13} although the methods are very similar. Previously we actually integrated along the path of outer turning points, i. e., the turning points on the dissociation side; when the curvature is negative this corresponds to a concave-side path. A second difference would occur for regions where $\kappa(s) > 1/u_>(s)$. In such regions the original Marcus-Coltrin path may have loops. In the present method, the effective mass is set equal to zero whenever $\kappa > 1/u_>(s)$, thereby effectively eliminating such loops. Such loops do not occur for the ground state for any system to which we have applied the MCP method in previous work or this paper, but they would occur for high-curvature systems.

E. Small-curvature method

Because of the great success enjoyed by the MCP method in numerical computations,^{4,5,7-10,12,13} its domain of validity was extended by the proposal of a new method, the small-curvature (SC) approximation.¹⁴ The SC method is an improvement over the MCP method in that it is designed to be applicable without singularities to large-curvature systems, although as implied by the name, and like the other methods presented in this paper, it is expected to be most accurate for small-curvature systems. When the curvature is large, i. e., $a(s) \geq 1$, the MCP is poorly defined, it may even have loops, and $\mu_{\text{eff}}^{\text{MCP}}(s, n)$ may be zero. Another pathology results when the ω' term in $\mu_{\text{eff}}^{\text{MCP}}(s, n)$ is large. Then

the MCP possesses a bubble and may be a longer tunneling path than the MEP. So whenever $a(s)$ or ω' is large the interpretation of the MCP as a minimum-imaginary-action tunneling path becomes suspect. We have shown¹⁴ that one can suppress the bubble effect and obtain an extended theory by the definition

$$\mu_{\text{eff}}^{\text{SC}}(s, 0) = \mu \min \left\{ \exp[-2\kappa(s)u_s(s) - [\kappa(s)u_s(s)]^2 + (u_s')^2], 1 \right\}, \quad (107)$$

which replaces $\mu_{\text{eff}}^{\text{MCP}}(s, 0)$ in Eq. (103). Equation (107) reproduces Eq. (102) to second order in $\kappa(s)u_s(s)$ while avoiding problems with divergences and correcting bubbles. We should emphasize that when the curvature is large, there is no guarantee that the unconstrained Schrödinger equation is valid and the SC should be regarded as an *ad hoc* extension to be justified numerically.

F. Second-order method

The methods presented in Subsecs. A and B neglected the coupling between the different vibrational channels. The method of Subsec. C included this effect implicitly. In the present subsection and the next we shall describe two ways to include this effect more explicitly. In this subsection we consider a second-order (SO) method that includes the diagonal component of the nonadiabatic coupling. There are two sources of nonadiabatic effects. The first is the centrifugal distortion of the vibrational motion; this effect was modeled by the choice of an energy-dependent vibrational basis in the DA model. The second contribution originates with the nonseparability of the Born-Oppenheimer potential in natural collision coordinates. The SO treatment goes beyond the DA method in that it includes this second source of nonadiabaticity.

The SO method, which was proposed by Gray *et al.*¹⁸ is a perturbative extension of the harmonic PA treatment of the classical Hamiltonian of Eq. (50). The unperturbed Hamiltonian is taken to be

$$H_C^0(p_s, s, n, q) = \frac{p_s^2}{2\mu} + V^{\text{MEP}}(s) + (n + \frac{1}{2})\hbar\omega(s). \quad (108)$$

Using classical perturbation theory²⁵ carried to second order, Gray *et al.*¹⁸ obtained the following energy conservation condition for the ground state:

$$E = \frac{p^2(s, E)}{2\mu} + V^{\text{MEP}}(s) + \epsilon_0(s) + G_A(s) + p^2(s, E)G_B(s) - p^4(s, E)G_C(s), \quad (109)$$

where

$$G_A(s) = \frac{\hbar^2}{64\mu} \left[\frac{\omega'}{\omega(s)} \right]^2, \quad (110)$$

$$G_B(s) = \frac{3a^2(s)}{4\mu} - \frac{\hbar}{16\mu^2} \left[\frac{\omega'}{\omega(s)} \right]^2 \frac{1}{\omega(s)}, \quad (111)$$

$$G_C(s) = \frac{1}{8} \frac{\kappa^2(s)}{\mu^3 \omega^2(s)} = \frac{a^2(s)}{8\mu^2 \hbar \omega(s)}. \quad (112)$$

Comparison of Eq. (109) to Eq. (38), using Eq. (43a), shows that

$$\mu_{\text{eff}}^{\text{SO}}(s, 0) = \mu / [1 + 2\mu G_B(s)], \quad (113)$$

$$V_{\text{ave}}^{\text{SO}}(s, 0) = V_a^G(s), \quad (114)$$

and

$$V_{\text{pert}}^{\text{SO}}(s, 0) = G_A(s) - G_C(s)p^4(s, E). \quad (115)$$

The classically forbidden region is defined in this method by

$$V_{\text{SO}}(s) > E, \quad (116)$$

where

$$V_{\text{SO}}(s) = V_a^{\text{SO}}(s, 0) + G_A(s). \quad (117)$$

In the classically forbidden region, Eq. (109) may be solved for an imaginary momentum, yielding

$$\text{Im } p(s, E) = \{2\mu_{\text{eff}}^{\text{SO}}(s, 0) [V_{\text{SO}}(s) - E]/G_{\text{SO}}(s, E)\}^{1/2}, \quad (118)$$

where

$$G_{\text{SO}}(s, E) = (1 + \{1 + 16[\mu_{\text{eff}}^{\text{SO}}(s, 0)]^2 G_C(s) [V_{\text{SO}}(s) - E]\}^{1/2})/2. \quad (119)$$

Reaction probabilities may be calculated from Eq. (118) in the usual way. This is called the SO ω' model. Neglect of the ω' terms in $G_A(s)$ and $G_B(s)$ gives an adiabatic model that we call SO.

Unlike the other methods considered in this paper, the SO ω' algorithm does not yield a zero value for $\theta(E = V_a^G)$. This reflects the presence of the term $G_A(s)$ in the potential. As a result the SO ω' TSAG procedure (where the SO ω' method is used only at tunneling energies as discussed in Sec. IV) yields a discontinuity for $T(E)$ at $E = V_a^G$. In practice however this discontinuity is so small for the systems considered that it does not seem particularly bothersome.

G. Semiclassical optical potential

The DA and SO methods described above represent two methodologies for the inclusion of centrifugal effects in the vibrational motion. In this section we introduce a third technique, the semiclassical-optical-potential (SOP) method, which has important advantages over the DA and SO methods. In particular, the SOP method is not restricted to the harmonic approximation or to a weak nonadiabatic-coupling approximation, as the DA and SO methods are.

The basis for the development of the SOP is the projected form of the Schrödinger equation first introduced by Feshbach²⁶:

$$\left[\text{PHP} + \text{PHQ} \frac{1}{E - \text{QH}Q} \text{QHP} - E \right] P\psi = 0. \quad (120)$$

We define the operator P to be a projection operator onto the ground vibration state, which is defined by Eq. (14) with $n=0$ [and neglecting the zero point curvature term of Eq. (34)], and Q is a projection operator onto the complementary space, i.e.,

$$Q = 1 - P. \quad (121)$$

When H is defined by Eq. (10) and nonadiabatic derivative terms [those containing κ' or $(\partial\phi_0/\partial s)_u$] are neglected, the vibrational matrix elements of the projected Hamiltonian quantities in Eq. (120) become

$$\langle \phi_0(u; s) | \text{PHP} | \phi_0(u; s) \rangle = -\frac{\hbar^2}{2\mu_{\text{eff}}(s, 0)} \frac{d^2}{ds^2} + V_a^G(s), \quad (122)$$

$$\langle \phi_i(u; s) | \text{QHP} | \phi_0(u; s) \rangle = -\frac{\hbar^2}{2\mu} A_{i0}(s) \frac{d^2}{ds^2}, \quad (123)$$

$$\langle \phi_0(u; s) | \text{PHQ} | \phi_i(u; s) \rangle = -\frac{\hbar^2}{2\mu} A_{0i}(s) \frac{d^2}{ds^2}, \quad (124)$$

and

$$\begin{aligned} \langle \phi_i(u; s) | \text{QH} | \phi_j(u; s) \rangle \\ = -\frac{\hbar^2}{2\mu} A_{ij}(s) \frac{d^2}{ds^2} + V_a(s, i) \delta_{ij}, \end{aligned} \quad (125)$$

where $\mu_{\text{eff}}(s, n)$ is defined by Eqs. (36) and (39). $A_{ij}(s)$ provides a generalization of $A_n(s)$:

$$A_{ij}(s) = \int_{-\infty}^{1/\kappa(s)} \chi_i(u; s) \frac{1}{\eta^2} \chi_j(u; s) du. \quad (126)$$

Notice that

$$A_{nn}(s) = \mu / \mu_{\text{eff}}(s, n). \quad (127)$$

As discussed in Sec. III, the integrals in $\mu_{\text{eff}}(s, n)$ and $A_{ij}(s)$ are singular in the absence of physical approximations. Here we eliminate the singularities associated with the multivalued regions of the coordinate system by expanding η^{-2} as in Eq. (59). This yields

$$\mu_{\text{eff}}(s, n) = \mu_{\text{eff}}^{\text{VA}}(s, n) \quad (128)$$

and

$$A_{ij}(s) = 2\kappa(s) \int_{-\infty}^{1/\kappa(s)} \chi_i^*(u; s) [u + \frac{3}{2}\kappa(s)u^2] \chi_j(u; s) du, \quad i \neq j. \quad (129)$$

The difficulty with using Eq. (120) is the computation of the resolvent, which is the inverse of a differential operator. To make this tractable we introduce the semiclassical approximation of replacing $(\hbar/i)\partial/\partial s$ by $p(s, E)$, as was done in Subsecs. C and E above. With this approximation the differential equation (120) is reduced to the following algebraic equation:

$$\begin{aligned} \frac{p^2(s, E)}{2\mu_{\text{eff}}^{\text{VA}}(s, 0)} + V_a^G(s) \\ + \sum_i' \sum_j' \frac{A_{0i}(s)A_{j0}(s) [p^2(s, E)/2\mu]^2}{[E - V_a(s, i)]\delta_{ij} - A_{ij}(s)p^2(s, E)/2\mu} - E = 0. \end{aligned} \quad (130)$$

Equation (130) can be solved for $p(s, E)$. For low energies $p(s, E)$ is imaginary, and it is used in Eq. (42) to compute the tunneling probabilities.

In general, the summation over i and j in Eq. (130) is a sum over a complete set, but here we make the approximation the sum contains only the single term $i=j=1$. In the harmonic approximation, Eq. (130) then yields

$$\text{Im } p(s, E) = \left(\frac{\alpha_+(s) - \left\{ \frac{[\alpha_-(s)]^2 + [2A_{01}(s)\mu_{\text{eff}}^{\text{VA}}(s, 1)\mu_{\text{eff}}^{\text{VA}}(s, 0)/\mu]^2 \Delta_{01}}{|A_{01}(s)|^2 \mu_{\text{eff}}^{\text{VA}}(s, 1)\mu_{\text{eff}}^{\text{VA}}(s, 0)/\mu^2 - 1} \right\}^{1/2}}{\alpha_-(s)} \right)^{1/2}, \quad (131a)$$

where

$$\alpha_{\pm}(s) = \Delta_1 \mu_{\text{eff}}^{\text{VA}}(s, 1) \pm \Delta_0 \mu_{\text{eff}}^{\text{VA}}(s, 0) \quad (131b)$$

and

$$\mu_{\text{eff}}^{\text{VA}}(s, 0) = \mu / [1 + \frac{3}{2}a^2(s)], \quad (132a)$$

$$\mu_{\text{eff}}^{\text{VA}}(s, 1) = \mu / [1 + \frac{9}{2}a^2(s)], \quad (132b)$$

$$|A_{01}(s)|^2 = 2a^2(s), \quad (133)$$

$$\Delta_n = E - V^{\text{MEP}}(s) - \epsilon_n(s). \quad (134)$$

We can put the SOP method into our general framework by comparing Eqs. (38) and (130); this yields

$$\mu_{\text{eff}}^{\text{SOP}}(s, 0) = \mu_{\text{eff}}^{\text{VA}}(s, 0), \quad (135)$$

$$V_{\text{ave}}^{\text{SOP}}(s, 0) = V_a^G(s), \quad (136)$$

and

$$V_{\text{pert}}^{\text{SOP}}(s, 0) = \frac{|A_{01}(s)|^2 \left[\frac{p^2(s, E)}{2\mu} \right]^2}{E - V_a(s, 1) - \frac{p^2(s, E)}{2\mu_{\text{eff}}^{\text{VA}}(s, 1)}}, \quad (137)$$

where $p(s, E)$ is imaginary and is given by Eq. (131). Notice that $V_{\text{pert}}^{\text{SOP}}(s, 0)$ is generally negative but it equals zero when $E = V^{\text{AG}}$.

As in the DA method the radicand of Eq. (131a) is more complicated than that of Eq. (44a), however, the turning points $s_{\pm}$ are obtained straightforwardly from

$$\Delta_0(s_{\pm}, E) = 0. \quad (138)$$

Also, for $E = V^{\text{AG}}$, the phase integral goes to zero and we can use the zero-curvature method, Eq. (46) for energies above V^{AG} .

H. Anharmonic generalizations

In this section we present anharmonic generalizations of the models considered in Subsecs. A–G. To avoid redundancy, we begin by pointing out two aspects of the generalization common to all the methods. Namely, we shall replace the harmonically calculated $\epsilon_n(t)$ with an anharmonic value calculated by the Morse I approximation⁹ when it appears in the adiabatic potential, and the quantity $a(s)$ [Eq. (8b)] will be calculated using the Morse oscillator turning point. The remaining generalizations are made on a case-by-case basis.

The anharmonic generalization of the PA method has been presented in an earlier paper.¹³ The lowest order effects of Morse anharmonicity were included in the effective reduced mass to yield, for the ground vibrational state

$$\mu_{\text{eff}}^{\text{PA}}(s, 0) = \mu [1 - \gamma^2(s)]^{3/2} \left[1 + \frac{3\kappa(s)}{a_M(s)k_M(s)} \right]^{-1}, \quad (139)$$

where $\gamma(s)$ is given by Eq. (55) and $a_M(s)$ and $k_M(s)$ are, respectively, the Morse range parameter and the reciprocal of the anharmonicity constant.¹³ Since the term $V_{\text{pert}}(s, 0)$ is zero in this method, no further generalization is necessary.

For the VA model, the anharmonic extension requires the evaluation of the expectation value of Eq. (60). Gallas²⁷ has presented an analytic formula for arbitrary moments of Morse oscillators. Using this formula, we obtain the following expression for the ground-state effective reduced mass:

$$\mu_{\text{eff}}^{\text{VA}}(s, 0) = \mu \left[1 + \frac{2\kappa(s)}{a_M(s)} \{ \ln k_M(s) - \psi[k_M(s) - 1] \} + \frac{3\kappa^2(s)}{a_M^2(s)} (\{ \ln k_M(s) - \psi[k_M(s) - 1] \}^2 + \psi'[k_M(s) - 1]) \right]^{-1}, \quad (140)$$

where ψ and ψ' are the digamma and trigamma functions.²⁸

The DA method has no analytic generalization for anharmonic vibrational potentials; this is a consequence of the use of the analytic properties of the harmonic potential to avoid the numerical iteration. For the present calculations, we apply the DA method to anharmonic systems by making the *ad hoc* replacement of $\omega(s)$ by $\epsilon_0(s)/2\hbar$.

Inclusion of anharmonicity in the MCP and SC methods is easy. To obtain the effective reduced mass we merely substitute the Morse turning point for the harmonic turning point in Eqs. (102) and (107).

For the SO method, like the DA method, there is no clear way to include anharmonicity. We suspect, however, that the integrals of classical perturbation theory can be treated similarly to those which occurred in the PA method.¹³ In the absence of a more rigorous procedure, the replacement of $\omega(s)$ by $\epsilon_0/2\hbar$ will be used here as was used previously.^{6(b)}

An important advantage of the SOP method is that anharmonicity is easily included. Since all the expectation values were done by the VA procedure, all that is necessary to convert to a Morse oscillator model is to use the Gallas formula. For convenience we make the further approximation of neglecting all but the first term ($i=j=1$) in the double sum of Eq. (130). Then the anharmonic result is obtained from Eq. (130) with the following replacements:

$$|A_{10}(s)|^2 = 4\kappa^2(s) [u_{01}^{(1)}(s)]^2, \quad (141)$$

$$\mu_{\text{eff}}^{\text{VA}}(s, 0) = \mu / [1 + 2\mu_{00}^{(1)}(s) \kappa(s) + 3\mu_{00}^{(2)}(s) \kappa^2(s)], \quad (142)$$

$$\mu_{\text{eff}}^{\text{VA}}(s, 1) = \mu / [1 + 2u_{11}^{(1)}(s) \kappa(s) + 3u_{11}^{(2)}(s) \kappa^2(s)], \quad (143)$$

$$u_{01}^{(1)}(s) = \frac{[k_M(s) - 3]^{1/2}}{a_M(s)} \{ \psi[k_M(s) - 1] - \psi[k_M(s) - 2] \}, \quad (144)$$

$$u_{00}^{(1)}(s) = \frac{1}{a_M(s)} \{ \ln k_M(s) - \psi[k_M(s) - 1] \}, \quad (145)$$

$$u_{00}^{(2)}(s) = \frac{1}{a_M^2(s)} \{ (\ln k_M(s) - \psi[k_M(s) - 1])^2 + \psi'[k_M(s) - 1] \}, \quad (146)$$

$$u_{11}^{(1)}(s) = \frac{k_M(s)}{a_M(s)} \left(\frac{k_M(s) - 2}{k_M(s) - 3} \{ \ln k_M(s) - \psi[k_M(s) - 3] \} - 2 \{ \ln k_M(s) - \psi[k_M(s) - 2] \} + \ln k_M(s) - \psi[k_M(s) - 1] \right), \quad (147)$$

$$u_{11}^{(2)}(s) = \frac{k_M(s) - 3}{a_M^2(s)} \left[\frac{k_M(s) - 2}{k_M(s) - 3} (\{ \psi[k_M(s) - 3] - \ln k_M(s) \}^2 + \psi'[k_M(s) - 3]) - 2(\{ \psi[k_M(s) - 2] - \ln k_M(s) \}^2 + \psi'[k_M(s) - 2]) + \{ \psi[k_M(s) - 1] - \ln k_M(s) \}^2 + \psi'[k_M(s) - 1] \right]. \quad (148)$$

VI. COMPARISON OF METHODS

Despite their differing physical justifications, the models presented in the previous section show a great deal of similarity. Each model can be recast in the general vibrationally adiabatic formalism of Eq. (38). The curvature effects are included by an effective reduced mass that accounts for the dependence of the reaction path length on the vibrational phase and by a distortion potential that includes the centrifugal effects on the vibrational motion. In the DA method, the distorting effect was shown to also be describable through an energy-dependent vibrational basis set.

As noted in the introduction, the seven models of Sec. IV are naturally divided into two categories. The first contains the PA, VA, DA, SO, and SOP methods, while the second includes the MCP and SC methods. We will now examine in detail the interrelationship between the theories using the harmonic results and low-order expansions in the parameter $a(s)$ of Eq. (8). Each model in the first category is motivated by the assumption that the vibrational wave function retains its general structure even at tunneling energies. For those models in this group, the general Eq. (38) is interpreted as an average of the full Schrödinger equation over the vibrational motion of the adiabatic coordinate. The effective reduced mass then reflects the net result of averaging the differential reaction path length over a vibrational distribution. From Eqs. (56), (62), (94), (113), and (135), we obtain for the harmonic ground-state case:

$$\mu_{\text{eff}}^{\text{PA}}(s, 0) = \mu [1 - \frac{3}{2} a^2(s) + \frac{3}{8} a^4(s) + \dots], \quad (149)$$

$$\mu_{\text{eff}}^{\text{VA}}(s, 0) = \mu [1 - \frac{3}{2} a^2(s) + \frac{9}{4} a^4(s) + \dots], \quad (150)$$

$$\mu_{\text{eff}}^{\text{DA}}(s, 0) = \mu [1 - \frac{3}{2} a^2(s)], \quad (151)$$

$$\mu_{\text{eff}}^{\text{SO}}(s, 0) = \mu [1 - \frac{3}{2} a^2(s) + \frac{9}{4} a^4(s) + \dots], \quad (152)$$

$$\mu_{\text{eff}}^{\text{SOP}}(s, 0) = \mu [1 - \frac{3}{2} a^2(s) + \frac{9}{4} a^4(s) + \dots]. \quad (153)$$

Thus, these methods all give the same effective reduced mass through terms of order $a^3(s)$. For all but the DA method, the derivation of the effective mass required the calculation of the ambiguous expectation value $A_0(s)$ of Eq. (36). The DA method is unique in that $A_0(s)$ cancels completely from the general equation. Thus, the agreement of the low-order DP result with effective reduced masses computed by the other methods provides further justification for the PA and VA procedures for doing the expectation values.

For the PA and VA models, the effective reduced mass is the only curvature effect included in the general equation. The DA, SO, and SOP methods also include centrifugal effects through $V_{\text{ave}}(s, n)$ or $V_{\text{pert}}(s, n)$. It is found that to low order in $a(s)$ all three yield, from Eqs. (98), (99), (114), (115), (136), and (137),

$$V_{\text{ave}}(s, 0) + V_{\text{pert}}(s, 0) - V_a^G(s) = -2a^2(s) [E - V_a^G(s)]^2 / \hbar \omega(s). \quad (154)$$

This term lowers the effective potential barrier at tunneling energies. It is encouraging that we have this agreement of the methods since at order $a^2(s)$ the SO and SOP methods are second order classical and quantum perturbation theories, respectively. Since the first two vibrational moments are identical in classical and quantum treatments for a harmonic oscillator, it is understandable that the theories agree. The DA method includes the centrifugal distortions in the vibrational basis without using formal perturbation theory. The agreement of the DA method with the SO and SOP methods reflects the fact that the distorted basis set of the DA method yields a separable theory up to order $a^2(s)$.

Of the models in the first category, the SOP method seems to be the most powerful. The SOP method, unlike the DA method, does not require artificial cutoffs as the energy is lowered; it is not restricted to classical vibrational motion as is the SO method; and it surpasses both the SO and DA methods in being easily extended to an anharmonic basis. Indeed, we think that the SOP method may have general utility, not restricted to tunneling calculations, for treating systems which satisfy the semiclassical assumptions.

The second category of models includes the SC and MCP methods. These methods compute the effective reduced mass as if the wave function collapses about the path of minimum action. The low-order results for $\mu_{\text{eff}}(s, 0)$ are smaller than those for theories of the first category; from Eqs. (102) and (107) we have

$$\mu_{\text{eff}}^{\text{MCP}}(s, 0) = \mu [1 - 2a(s) + a^2(s) + (u'_s)^2], \quad (155)$$

$$\mu_{\text{eff}}^{\text{SC}}(s, 0) = \mu \left\{ 1 - 2a(s) + a^2(s) + \frac{2}{3}a^3(s) - \frac{5}{6}a^4(s) + \dots + (u'_s)^2 [1 - 2a(s) + \dots] \right\}. \quad (156)$$

This replacement of the averaging model of the first category of methods by the path-integral model of the second makes it unclear how to include the centrifugal distortions. Thus, for both the MCP and SC methods the V_{pert} term is set to zero.

A final point of comparison concerns the ω' terms in the $\text{PA}\omega'$ and $\text{SO}\omega'$ approximations. It is clear that

these terms are related to the nonadiabatic derivative terms discarded in Sec. III. However, inspection of Eqs. (27), (53), and (109) reveals a difference in the numerical factor multiplying the quantum and classical results even in the zero-curvature approximation. This difference reflects the inequality (between classical and quantum methods) for u^4 moments which are needed for these terms. To gauge the effects of this potential non-separability, the SO calculations will be done both with and without the ω' terms. We shall find that for the cases considered the effects are very small.

VII. NUMERICAL RESULTS

In this section we present numerical results of the methods considered in the last section for the three collinear reactions $\text{H} + \text{H}_2$, $\text{D} + \text{D}_2$, and $\text{D} + \text{H}_2$ on the potential energy surface of one of the authors and Kuppermann²⁰ and for the collinear reaction $\text{Cl} + \text{D}_2$ on the potential energy surface of Stern, Persky, and Klein (SPK; we use their final surface which they denote GSW²⁹). These systems are representative protium and deuterium transfers with medium reaction-path curvature. Two levels of numerical results are presented: ground-state reaction probabilities as functions of energy and thermally averaged transmission coefficients.

The ground-state reaction probabilities $T(E)$ are computed from Eqs. (29), (44), and (45) and are compared to accurate quantal results obtained in previous work^{20,30,31} by solving the full two-mathematical-dimensional Schrödinger equation. These results are given in Tables I–IV.

The semiclassical adiabatic ground-state (SAG) transmission coefficients are computed using Eq. (48). The accurate transmission coefficients are computed as the ratio of the accurate two-mathematical-dimensional quantal rate constants^{20,30,32} to the neglect-of-tunneling vibrationally adiabatic rate constants $k^A(T)$,⁹ where, as discussed previously,⁹ $k^A(T)$ is computed with quantized $\epsilon_n(s)$ but with classical reaction-coordinate motion and neglect of reaction-path curvature. The thermally averaged transmission coefficients are given in Tables V–VIII.

In addition to the seven methods discussed in Sec. V

TABLE I. Ground-state reaction probabilities for collinear $\text{H} + \text{H}_2$.

E^a (kcal/mol)	Accurate ^b	SAG								
		MEP	PA	VA	DA	SO	SO ω'	SOP	MCP	SC
6.514	3.9(−7) ^c	1.0(−9)	1.1(−8)	6.7(−9)	3.3(−8)	9.8(−9)	9.5(−9)	1.4(−8)	1.8(−7)	1.7(−7)
7.289	4.2(−6)	1.6(−8)	1.6(−7)	9.7(−8)	3.8(−7)	1.3(−7)	1.2(−7)	1.8(−7)	2.2(−6)	1.9(−6)
8.158	2.8(−5)	2.6(−7)	2.1(−6)	1.4(−6)	4.2(−6)	1.6(−6)	1.6(−6)	2.2(−6)	2.4(−5)	2.1(−5)
9.000	1.8(−4)	3.5(−6)	2.2(−5)	1.5(−5)	3.8(−5)	1.7(−5)	1.6(−5)	2.2(−5)	2.0(−4)	1.7(−4)
9.918	1.5(−3)	5.5(−5)	2.7(−4)	1.9(−4)	4.0(−4)	2.0(−4)	1.9(−4)	2.4(−4)	1.8(−3)	1.6(−3)
10.859	1.5(−2)	9.5(−4)	3.4(−3)	2.6(−3)	4.3(−3)	2.5(−3)	2.5(−3)	2.9(−3)	1.5(−2)	1.3(−2)
11.737	1.3(−1)	1.6(−2)	3.7(−2)	3.1(−2)	4.2(−2)	3.0(−2)	2.9(−2)	3.2(−2)	9.4(−2)	8.8(−2)
12.554	5.9(−1)	3.1(−1)	3.6(−1)	3.5(−1)	3.7(−1)	3.4(−1)	3.4(−1)	3.5(−1)	4.1(−1)	4.1(−1)
13.396	9.7(−1)	9.7(−1)	9.4(−1)	9.5(−1)	9.3(−1)	9.5(−1)	9.5(−1)	9.5(−1)	8.6(−1)	8.7(−1)

^aTotal energy including reactant zero point energy.

^cNumbers in parentheses are powers of 10.

^bReference 20.

TABLE II. Ground-state reaction probabilities for collinear D+D₂.

E ^a (kcal/mol)	Accurate ^b	SAG								
		MEP	PA	VA	DA	SO	SO ω'	SOP	MCP	SC
6.904	2.6(-7) ^c	2.1(-9)	1.8(-8)	1.2(-8)	5.9(-8)	2.0(-8)	2.0(-8)	2.9(-8)	4.8(-7)	4.1(-7)
7.531	2.1(-6)	2.8(-8)	2.0(-7)	1.4(-7)	5.2(-7)	2.1(-7)	2.0(-7)	2.9(-7)	4.3(-6)	3.7(-6)
8.167	1.7(-5)	3.6(-7)	2.2(-6)	1.6(-6)	4.5(-6)	2.1(-6)	2.1(-6)	2.9(-6)	3.6(-5)	3.1(-5)
9.000	2.4(-4)	9.8(-6)	4.7(-5)	3.6(-5)	7.5(-5)	4.2(-5)	4.1(-5)	5.3(-5)	5.0(-4)	4.4(-4)
9.915	4.4(-3)	3.7(-4)	1.3(-3)	1.0(-3)	1.6(-3)	1.1(-3)	1.0(-3)	1.2(-3)	7.7(-3)	7.0(-3)
10.863	7.5(-2)	1.7(-2)	3.6(-2)	3.1(-2)	3.9(-2)	3.1(-2)	3.0(-2)	3.3(-2)	9.7(-2)	9.1(-2)
11.786	6.4(-1)	4.7(-1)	4.8(-1)	4.7(-1)	4.8(-1)	4.7(-1)	4.7(-1)	4.7(-1)	4.8(-1)	4.8(-1)
12.238	9.1(-1)	8.4(-1)	7.9(-1)	8.0(-1)	7.9(-1)	8.0(-1)	8.0(-1)	8.0(-1)	7.0(-1)	7.1(-1)

^aTotal energy including reactant zero point energy.^cNumbers in parentheses are powers of 10.^bReference 29.TABLE III. Ground-state reaction probabilities for collinear D+H₂.

E ^a (kcal/mol)	Accurate ^b	SAG								
		MEP	PA	VA	DA	SO	SO ω'	SOP	MCP	SC
6.664	6.4(-7) ^c	7.2(-10)	1.1(-8)	5.4(-9)	3.5(-8)	8.1(-9)	7.8(-9)	1.2(-8)	1.6(-7)	1.4(-7)
7.633	7.6(-6)	2.6(-8)	3.0(-7)	1.6(-7)	7.9(-7)	2.2(-7)	2.1(-7)	3.0(-7)	3.7(-6)	3.2(-6)
9.408	6.4(-4)	1.1(-5)	7.4(-5)	4.6(-5)	1.3(-4)	5.0(-5)	4.9(-5)	6.2(-5)	5.1(-4)	4.6(-4)
10.377	8.0(-3)	2.9(-4)	1.3(-3)	9.1(-4)	2.0(-3)	9.2(-4)	8.9(-4)	1.1(-3)	6.1(-3)	5.6(-3)
11.129	5.7(-2)	4.0(-3)	1.3(-2)	9.5(-3)	1.6(-2)	9.1(-3)	9.0(-3)	1.0(-2)	3.7(-2)	3.5(-2)
11.881	3.3(-1)	8.1(-2)	1.3(-1)	1.1(-1)	1.4(-1)	1.1(-1)	1.1(-1)	1.2(-1)	2.0(-1)	1.9(-1)
12.632	8.3(-1)	4.8(-1)	4.8(-1)	4.8(-1)	4.8(-1)	4.8(-1)	4.8(-1)	4.8(-1)	4.9(-1)	4.9(-1)
13.384	9.6(-1)	8.9(-1)	8.4(-1)	8.6(-1)	8.4(-1)	8.6(-1)	8.6(-1)	8.6(-1)	7.8(-1)	7.8(-1)

^aTotal energy including reactant zero point energy.^bReference 29.^cNumbers in parentheses are powers of 10.TABLE IV. Ground-state reaction probabilities for collinear Cl+D₂.

E ^a (kcal/mol)	Accurate ^b	SAG								
		MEP	PA	VA	DA	SO	SO ω'	SOP	MCP	SC
7.003	8.8(-5) ^c	1.0(-6)	9.3(-6)	4.5(-6)	2.0(-5)	5.3(-6)	5.2(-6)	6.3(-6)	5.0(-5)	5.1(-5)
7.350	1.2(-3)	2.3(-5)	1.6(-4)	8.4(-5)	3.0(-4)	9.3(-5)	9.1(-5)	1.1(-4)	7.0(-4)	7.1(-4)
7.696	1.0(-2)	3.2(-4)	1.7(-3)	9.7(-4)	2.8(-3)	1.0(-3)	1.0(-3)	1.1(-3)	5.9(-3)	6.0(-3)
8.157	1.0(-1)	6.8(-3)	2.3(-2)	1.5(-2)	3.2(-2)	1.6(-2)	1.5(-2)	1.7(-2)	5.6(-2)	5.6(-2)
8.849	6.5(-1)	2.7(-1)	3.2(-1)	3.1(-1)	3.3(-1)	3.1(-1)	3.1(-1)	3.1(-1)	3.8(-1)	3.8(-1)
9.310	9.8(-1)	7.2(-1)	6.7(-1)	6.9(-1)	6.6(-1)	6.9(-1)	6.9(-1)	6.9(-1)	6.1(-1)	6.2(-1)

^aTotal energy including reactant zero point energy.^bReference 30.^cNumbers in parentheses are powers of 10.TABLE V. Transmission coefficients for collinear H+H₂.

T (K)	Accurate ^a	SAG												
		MEP	PAT	VAT	DAT	DA	SOT	SO ω' T	SOPT	SOP	MCPT	SCT	SCda	SC
200	29.8	2.08	4.77	3.82	6.63	6.58	3.93	3.87	4.51	4.48	24.5	21.8	21.7	21.7
300	3.32	1.25	1.61	1.50	1.72	1.67	1.48	1.48	1.53	1.50	2.88	2.72	2.67	2.63
400	2.01	1.12	1.29	1.24	1.33	1.29	1.23	1.23	1.25	1.22	1.75	1.70	1.66	1.61
600	1.44	1.05	1.12	1.10	1.14	1.11	1.10	1.10	1.11	1.09	1.31	1.29	1.25	1.22
1000	1.20	1.02	1.05	1.04	1.06	1.04	1.04	1.04	1.04	1.03	1.13	1.12	1.10	1.07

^aReference 9.

TABLE VI. Transmission coefficients for collinear D+D₂.

T (K)	Accurate ^a	SAG												
		MEP	PAT	VAT	DAT	DA	SOT	SOω'T	SOPT	SOP	MCPT	SCT	SCda	SC
200	6.50	2.02	3.02	2.76	3.36	3.33	2.79	2.77	2.96	2.94	9.00	8.29	8.26	8.21
300	2.30	1.33	1.57	1.52	1.62	1.59	1.51	1.51	1.54	1.52	2.46	2.37	2.33	2.29
400	1.69	1.17	1.29	1.27	1.31	1.28	1.26	1.26	1.28	1.26	1.68	1.64	1.61	1.57
600	1.32	1.07	1.13	1.12	1.14	1.11	1.11	1.11	1.12	1.10	1.29	1.28	1.25	1.22
1000	1.13	1.02	1.05	1.05	1.06	1.04	1.04	1.04	1.05	1.04	1.12	1.12	1.10	1.07

^aReference 9.TABLE VII. Transmission coefficients for collinear D+H₂.

T (K)	Accurate ^a	SAG												
		MEP	PAT	VAT	DAT	DA	SOT	SOω'T	SOPT	SOP	MCPT	SCT	SCda	SC
200	39.2	3.16	6.49	5.18	8.86	8.83	5.24	5.16	5.78	5.73	23.8	21.9	21.8	21.8
300	5.15	1.61	2.08	1.92	2.25	2.22	1.90	1.89	1.95	1.93	3.40	3.28	3.25	3.22
400	2.89	1.30	1.51	1.44	1.57	1.54	1.43	1.43	1.45	1.43	1.99	1.94	1.91	1.88
600	1.81	1.12	1.21	1.18	1.23	1.21	1.18	1.18	1.19	1.17	1.39	1.37	1.35	1.32
1000	1.37	1.04	1.08	1.07	1.09	1.07	1.07	1.07	1.07	1.06	1.15	1.15	1.13	1.10

^aReference 9.TABLE VIII. Transmission coefficients for collinear Cl+D₂.

T (K)	Accurate ^a	SAG												
		MEP	PAT	VAT	DAT	DA	SOT	SOω'T	SOPT	SOP	MCPT	SCT	SCda	SC
200	4.56	1.58	2.08	1.88	2.30	2.26	1.88	1.87	1.91	1.89	2.99	2.97	2.93	2.89
300	2.46	1.23	1.41	1.34	1.48	1.44	1.34	1.34	1.35	1.33	1.72	1.71	1.67	1.63
400	1.87	1.12	1.23	1.19	1.27	1.23	1.19	1.19	1.19	1.17	1.40	1.39	1.35	1.32
600	1.48	1.05	1.11	1.09	1.13	1.10	1.09	1.09	1.09	1.07	1.19	1.19	1.16	1.14
1000	1.18	1.02	1.05	1.04	1.05	1.03	1.04	1.04	1.04	1.03	1.09	1.08	1.06	1.05

^aReferences 9 and 31.

we consider another, which may be called the minimum-energy-path (MEP) or zero-curvature (ZC) approximation.^{13,20} In this method we use Eqs. (46a)–(46c) for both tunneling and nonclassical reflection. This method assumes that the tunneling path is centered on the MEP, which would strictly be true only for zero curvature ($\kappa = 0$).

Finally, Table IX gives the percent contributions to the tunneling-corrected vibrationally adiabatic rate constants. To compute these contributions we divide the transmission coefficient into two parts

$$\kappa(T) = \kappa_{\text{tun}}(T) + \kappa_{\text{ob}}(T), \quad (157)$$

where the tunneling contribution is

$$\kappa_{\text{tun}}(T) = \langle T_C \rangle^{-1} \int_{\epsilon_0^{\text{max}}}^{\nu^{\text{AG}}} T(E) e^{-\beta E} dE \quad (158)$$

and the over-barrier contribution is

$$\kappa_{\text{ob}}(T) = \langle T_C \rangle^{-1} \int_{\nu^{\text{AG}}}^{\infty} T(E) e^{-\beta E} dE. \quad (159)$$

Then the fractional tunneling contribution to the ground-state reaction rate is $\kappa_{\text{tun}}(T)/[\kappa_{\text{tun}}(T) + \kappa_{\text{ob}}(T)]$.

VIII. DISCUSSION OF NUMERICAL RESULTS

The purpose of Table IX is simply to illustrate that a very large fraction of the calculated thermal rate constants is due to the tunneling process, even at temperatures as high as 400–600 K, and that all methods for all four reactions yield fractional tunneling contributions greater than one-half (actually 0.52–0.97) at 200 K. The recognition of the large quantitative role of tunneling in many reactions is one of the main motivations for the present attempt to understand the systematics and accuracy of methods for treating this process.

Next consider the low-energy reaction probabilities. The MEP method underestimates these by over two orders of magnitude for all three isotopic cases of H + H₂ and by almost two orders of magnitude for Cl + D₂. The PA-type methods (PA, VA, DA, SO, SOω', and

TABLE IX. Tunneling contributions in the ground-state reaction rate for collinear reactions.

T (K)	Percent tunneling				
	MEPSAG	DATSAG	DASAG	SCTSAG	SCdaSAG
$\text{H} + \text{H}_2$					
200	61	88	88	96.3	96.5
300	32	50	51	69	70
400	21	33	35	48	49
600	13	20	20	29	29
$\text{D} + \text{D}_2$					
200	63	78	78	91.	91
300	40	50	51	66	67
400	28	36	37	49	50
600	18	22	23	31	31
$\text{D} + \text{H}_2$					
200	78	92	92	96.8	96.9
300	53	66	67	77	77
400	39	49	50	59	60
600	24	31	32	38	39
$\text{Cl} + \text{D}_2$					
200	52	67	68	74	75
300	33	45	46	52	53
400	24	33	34	39	40
600	15	21	21	25	26

SOP) reduce the largest errors by about one order of magnitude; thus they still have errors of over an order of magnitude for the cases in Tables I–III. In contrast, the MCP-type methods (MCP and SC) are still accurate within about a factor of 2 ($\text{H} + \text{H}_2$, $\text{D} + \text{D}_2$, and $\text{Cl} + \text{D}_2$) to 4 ($\text{D} + \text{H}_2$) even at the lowest energy considered. As the energy is raised all the methods become more accurate but the MCP-type methods remain more accurate than the PA-type methods until the accurate $T(E)$ equals at least 0.6 for $\text{H} + \text{H}_2$, $\text{D} + \text{D}_2$, and $\text{Cl} + \text{D}_2$, and until it equals at least 0.8 for $\text{D} + \text{H}_2$. In all three cases, the MCP-type methods begin to overestimate the nonclassical reflection when the approximate $T(E)$ values exceed 0.5.

It is also interesting to compare the PA-type methods to one another. The fact that these methods generally agree reasonably well indicates that they do indeed incorporate the same physical effects. That may not have been too obvious from their original derivations but the discussion of Sec. VI already hinted that this would be the case. In fact, one of the main points of this paper is that all PA-type methods underestimate the tunneling and no significant improvement can be expected from further refinements of this kind of approach. The three methods DA, SO, and SOP that include the physical effects of virtual coupling to states differing from the reference adiabatic state used for the averaged potential do yield more accurate results than the PA and VA methods, but they are still significantly less accurate than the MCP and SC methods.

The good agreement of the SO and SOw' results with

each other is a numerical confirmation of the arguments we made about nonadiabatic derivative terms in Sec. III.

The MCP and SC approximations are also similar to each other, a result we have discussed previously.^{5,14} The significant improvement offered by these methods as compared to the PA-type methods indicates clearly that the leading term in $\mu_{\text{eff}}(s, 0) - \mu$ is of order $a(s)$ rather than $a^2(s)$.

The consequences of these trends for thermally averaged transmission coefficients are clearly exemplified in Tables V–VIII. Here we also compare the results of the DATSAG to the DASAG method, the SOPSAG to the SOPSAG method, and the SCTSAG to the SCdaSAG method. In each case these pairs differ only in the method used to treat nonclassical reflection. The fact that the differences within these pairs are small justifies the choice of a simple treatment of nonclassical reflection. Of the MEP and da procedures for nonclassical reflection, the latter represents a more realistic physical model. The former neglects curvature in the classically allowed region and there is no reason to believe that curvature will be an unimportant ingredient in the dynamics there. For the methods of choice one might prefer the SCTSAG, MCPTSAG, SCdaSAG, and MCPdaSAG methods to the SCSAG and MCPSAG versions because there is no good physical motivation to base the treatment of nonclassical reflection on a model involving collapse of the wave function about a preferred path. The fact that these six methods yield similar thermally averaged results for all cases studied here does, however, provide a numerical justification for the way the SCSAG and MCPSAG methods have been used to mathematically extrapolate the tunneling results into the nonclassical-reflection region for the calculation of thermally averaged rate constants. Since the transmission coefficients predicted by the SCTSAG and SCdaSAG methods are very similar, we will concentrate the rest of our discussion on the simpler SCTSAG version. There is no physical motivation to restrict the inclusion of curvature effects to the tunneling region for the PA-type methods since the physical model underlying these methods is valid both below and above threshold. In the present paper, however, we present the results for the PATSAG, VATSAG, and SOTSAG methods rather than the PASAG, VASAG, and SOSAG in order to make the comparisons between the various calculations directly reflect only the differences in the tunneling region, which is our main focus.

Tables V–VIII show that the PA-type methods are significantly less accurate than the MCP and SC methods for including reaction-path curvature effects in thermally averaged calculations. The difference between the PA-type and MCP-type methods is particularly striking for $\text{H} + \text{H}_2$ and $\text{D} + \text{D}_2$ at 300 and 400 K. For example for $\text{D} + \text{D}_2$ at 400 K the MEP result is 1.2, the PA-type methods yield about 1.3, the MCPTSAG and SCTSAG methods yield about 1.65, and the accurate answer is 1.7. Thus the PA-type methods make up about one-quarter of the error in the MEPSAG calculation, but the MCPTSAG and SCTSAG methods correct about three-quarters of the error.

TABLE X. Transmission coefficients for three-dimensional $H+H_2$ on the LSTH potential energy surface.

T (K)	Empirical	SAG							
		MEP	PAT	PA	VAT	DAT	DA	SOT	SO ω 'T
300	11.0	1.83	2.92	2.87	2.56	3.34	3.29	2.54	2.53
444	2.9	1.29	1.61	1.57	1.51	1.71	1.66	1.50	1.50
549	2.3	1.18	1.37	1.34	1.31	1.43	1.39	1.31	1.30
1000		1.05	1.12	1.09	1.10	1.14	1.10	1.09	1.09

T (K)	Empirical	SAG						
		SOPT	SOP	MCPT	MCP	SCT	SCda	SC
300	11.0	2.78	2.75	6.76	6.66	6.16	6.11	6.06
444	2.9	1.56	1.52	2.46	2.36	2.34	2.29	2.24
549	2.3	1.34	1.31	1.85	1.75	1.78	1.73	1.69
1000		1.10	1.08	1.26	1.19	1.24	1.20	1.17

IX. MULTIDIMENSIONAL REACTIONS

The MEPSAG, MCPSAG, PASAG, and SCSAG methods have been applied to three-dimensional atom-diatom reactions with collinear minimum-energy paths in several previous papers.^{5,10,12,13,33-36} The MEPSAG, SOSAG, and SCSAG methods have also been applied to a three-dimensional diatom-diatom example with a noncollinear minimum-energy path.^{5,6(b)} In this section, we present a few new results with the methods considered in this article for three-dimensional atom-diatom reactions.

For atom-diatom reactions with collinear minimum-energy paths the generalization of the methods considered here becomes very simple. The additional vibrational degrees of freedom are a doubly degenerate bend, and there is no component of reaction-coordinate curvature¹¹ in these degrees of freedom. Thus the reaction coordinate remains localized in two degrees of freedom, and bending degrees of freedom only contribute through their zero point energy. We must replace $\epsilon_0(s)$ in Eq. (28a) by the sum of the stretching and bending zero point energies. For the calculations presented here the bending zero point energies were all estimated by the same methods as we used in Refs. 12 and 13.

Table X shows some results we obtained for the three-dimensional $H+H_2$ reaction for the accurate (LSTH)³⁷ potential energy surface. (We consider the distinguishable-atom rate constant; this is easily related to the experimental observables by proper consideration of the nuclear spin statistics.^{10,38,39}) Comparison of the trends in this table to those in Table V indicates that the relative amounts of tunneling predicted by the various methods in three dimensions differ little from the collinear case. The second column of Table X is labeled empirical; we computed this as the value that would be required to make full transition-state theory calculations¹⁰ agree with experiment.^{40,41} We see that the MCPTSAG, SCSAG, SCdaSAG, and SCTSAG all yield reasonably good agreement with the empirical values. This comparison and also a comparison¹⁹ to the accurate three-dimensional quantal results³⁹ for the less accurate potential surface no. 2 of Porter and Karplus (PK2)⁴² give no reason to believe that the methods discussed here are any less accurate for three-dimensional reactions than for collinear reactions. Thus, in order

to assess the relative importance of tunneling and non-tunneling contributions in three dimensions, we use the SCTSAG method for several three-dimensional reactions with potential energy surfaces from the literature. In addition to potential surfaces already discussed we used the Muckerman no. 5 potential energy surface (M5)⁴³ for FH_2 and the potential surface of Raff *et al.* (RSPTS)⁴⁴ for $I+H_2$.

The results for 15 three-dimensional examples, nine H transfers, and six D transfers, are given in Table XI. In every case the table shows that the fraction of ground-state reactions occurring by the tunneling route is higher in three dimensions than in the collinear case. This is due in part to the higher vibrationally adiabatic barrier heights for this class of reaction in three dimensions than in one (this occurs for any reaction with a collinear minimum-energy path since the bending zero point energy is zero for separated reactants). We conclude that tunneling is important in reactions like this, and it makes at least a significant contribution, and in some cases the dominant contribution, to thermal rate constants for three-dimensional reactions.

TABLE XI. Percentage of ground-state reaction that occurs by tunneling as computed by SCTSAG method.

System (surface)	Collinear 300 K	3D			
		200 K	300 K	400 K	600 K
$H+H_2$ (PK2)	86	99.7	95.1	85	64
$Mu+H_2$ (PK2)	84	99.90	96.3	86	63
$Mu+D_2$ (PK2)	68	98.4	85	67	44
$H+H_2$ (LSTH)	69	99.0	88	71	48
$T+HD$ (LSTH)	65	94	74	57	37
$Mu+H_2$ (LSTH)	85	99.91	95.2	82	57
$Mu+D_2$ (LSTH)	66	97.7	79	61	39
$Cl+H_2$ (SPK)	47	92	73	58	38
$Cl+D_2$ (SPK)	52	87	66	50	33
$Cl+T_2$ (SPK)	53	84	62	47	31
$Cl+HD$ (SPK)	55	84	63	48	32
$Cl+DH$ (SPK)	52	89	67	50	32
$F+H_2$ (M5)	13	42	26	19	12
$F+D_2$ (M5)	11	29	18	13	8
$I+H_2$ (RSPTS)	0	19	12	9	6

X. CONCLUSION

In the Introduction we set forth three goals for this research. We now review our progress towards their attainment. The first goal was to present a unified formalism capable of including a wide range of particular one-dimensional representations of the reaction process. We chose as the foundation of this formalism vibrational adiabaticity in natural collision coordinates. In Sec. II it was shown that a standard adiabatic treatment of the system wave function was justified only in the small-curvature case as defined by Eq. (7). An adiabatic treatment can be extended in an *ad hoc* fashion by the small-curvature (SC) approximation to systems of intermediate curvature (defined as when the radius of curvature is approximately equal to the range of the vibrational restoring potential). However, for systems with large curvature we expect that a radical departure from the adiabatic method will be necessary. In Sec. III we used the adiabatic method and natural collision coordinates to define a general one-dimensional Schrödinger equation for the reaction-coordinate motion. This equation goes beyond the zero-curvature approximation in two ways; first in the presence of the effective reduced mass $\mu_{\text{eff}}(t, n)$, which accounts for the vibrational-phase dependence of the reaction path length, and second through the nonadiabatic coupling potential $V_{\text{pert}}(t, n)$. In Sec. V we made special choices of the collision coordinates, the adiabatic separation of the Hamiltonian, and the effective reduced mass, thereby obtaining seven different models as special cases of the general equation. Each of the choices was justified on physical grounds. For all seven methods we neglected the nonadiabatic derivative terms and the centripetal zero point terms to make the relationship between a physical perspective and the unspecified quantities of the one-dimensional model more transparent. If one retained these terms, especially the derivative-of-curvature terms, this relationship would be masked and the one-dimensional treatment would be untenable.

The second goal of this paper was to use the general formalism to develop and discuss various possible approaches to obtaining one-dimensional approximation schemes for multidimensional tunneling problems. We have repeatedly stressed that there are two alternative approaches for reactive tunneling. One view is similar to Born–Oppenheimer-type adiabaticity. Here one assumes that an effective one-dimensional equation of motion is obtained by averaging over the rapid vibrational motion. The second interpretation is more closely related to the path-integral representation of quantum mechanics. There one has a dominant, minimum-action path and the semiclassical reaction probability is found through the action integral along this path. The essential difference between the two approaches lies in how the expectation value defining the effective reduced mass $\mu_{\text{eff}}(t, n)$ is performed. Since the general formalism of Sec. III is derived from a close-coupling-type argument, the first interpretation is included very naturally. There the expectation value for the effective reduced mass is found from a vibrational average of the curvature prefactor in the reaction-coordinate kinetic energy operator at each value of the

reaction-coordinate t . It was shown in Sec. III that the strict mathematical definition of the effective reduced mass Eq. (36) led to ambiguities which could not be formally resolved. But on physical grounds it was shown that the phase-average (PA) or vibrational-average (VA) prescription for $\mu_{\text{eff}}(t, n)$ was sensible in the case of small curvature. Furthermore the dynamical-path vibrational-average (DA) method, which is unique in not requiring the explicit calculation of $\mu_{\text{eff}}(t, n)$, yields the same low-order harmonic result as the PA and VA methods. This provides a better justification for the validity of these methods. The second-order (SO) and semiclassical-optical-potential (SOP) methods involve the PA and VA choices for $\mu_{\text{eff}}(t, n)$, respectively, but the SOP method is distinguished by its generality and the ease with which quantum mechanical anharmonicity effects can be included.

In the path-integral view $\mu_{\text{eff}}(t, n)$ is interpreted as a metric factor relating the differential length of the reaction coordinate to that of the dominant tunneling path. The reference coordinate system (v, t) is used to make an adiabatic separation, thus eliminating the problem of evaluating a multidimensional root trajectory. The path-integral methods lead to significantly different values of $\mu_{\text{eff}}(t, n)$ than the vibrational-averaging methods do, and this causes large numerical differences between the tunneling probabilities calculated by the two approaches. It was our original hope that the vibrational-average interpretations could be brought into good agreement with the exact tunneling results if a sufficiently realistic dynamical model were used. To this end, the DA, SO, and SOP methods were introduced to model the centrifugal distortions. It was found, however, that these methods still could not compete with the Marcus–Coltrin-path (MCP) and small-curvature (SC) versions of the path-integral approach. However, we do feel that the SOP method in particular may provide a good description for the classically allowed regime. In this regime the vibrational wave function does not collapse around a dominant path. Further work applying the SOP method with exact one-dimensional quantum mechanics for the reaction-coordinate motion would be very interesting.

The final goal for this paper was to provide sufficient numerical testing to come to a conclusion concerning the best physical model for reactive tunneling in small- and intermediate-curvature systems. The numerical results of Sec. VI leave little doubt as to the methods of preference. The MCP and SC methods consistently yield the best results. Thus the concept of vibrational collapse about a minimum-action path leads to more accurate results than the preservation of the vibrational averaging in the tunneling region.

We have used the final recommended method of this paper, the small-curvature-tunneling semiclassical adiabatic ground-state (SCTSAG) method, to compute the fraction of reactive events that occur by tunneling for several ground-state reactions involving H-atom and D-atom transfer in both 1D and 3D. We find these fractions to be very large; for example for nine 3D cases involving H transfer the average tunneling fraction

is 0.81, 0.69, and 0.57 at 200, 300, and 400 K, respectively.

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APPENDIX

To illustrate how the classical phase average method is extended to derive Eqs. (26) and (27), consider

$$\begin{aligned} I &= \int \eta^{-3} u |\chi_0|^2 du \\ &= \int \eta^{-3} u V_R^{-1}(u) du / \int V_R^{-1} du \\ &= \frac{3}{2} \kappa \alpha^{-2} (1 - \alpha^2)^{-5/2}, \end{aligned} \quad (\text{A1})$$

where $V_R(u)$ is the classical speed $[2(\epsilon_0 - \frac{1}{2}\mu\omega^2 u^2)/\mu]^{1/2}$. Alternatively

$$I = \frac{1}{2} \frac{\partial}{\partial \kappa} A_0^{\text{PA}}, \quad (\text{A2})$$

which gives the same result. Similarly

$$\int \frac{u}{\eta^3} \chi_0 \chi_0' du = \frac{\alpha'}{4} \frac{\partial}{\partial \kappa} \frac{\partial}{\partial \alpha} A_0^{\text{PA}} \quad (\text{A3})$$

and so forth.

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